Manifold-Aware Concept Erasure

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Abstract

Concept erasure aims to remove a target concept from a representation with minimal damage to the rest of the representation, yet unconstrained updates in ambient space can distort unrelated concepts. We propose TanGCE: a manifold-aware erasure method that estimates a local tangent space at each representation via the SVD of the secants to its nearest neighbors, and restricts every erasure step to lie inside that locally estimated subspace. The intuition is that if representations concentrate near a low-dimensional neighborhood, effective edits should stay on that local support rather than push them off it. To exhaust easy concept signal before the iterative loop runs, we precede it with two closed-form moment-matching projections that remove the rank-1 mean-shift between classes (LEACE) and the rank-2 covariance asymmetry between classes; the iterative tangent-constrained loop then erases the residual nonlinear concept signal. Across evaluations spanning 13 language models from five families (0.5B–27B parameters) and a cross-modal vision benchmark (CelebA with CLIP ViT-B/32), the full pipeline strictly beats the strongest published unsupervised baseline (Obliviator) at every control-damage budget on the 39-panel language-model grid, reaching the majority-class floor on $23/39$ panels (vs. $14/39$ for Obliviator).

1 Introduction

Effective concept erasure must do two things at once: remove the target concept and preserve the rest of the representation. This is especially difficult for nonlinear concepts: even after adversarial training drives an in-training adversary to chance accuracy, a post-hoc classifier trained on the same encoded representations can still recover the protected attribute, as shown by Elazar and Goldberg [Elazar and Goldberg, 2018].

Concerns about demographic bias in learned representations have motivated methods for identifying, interpreting, and removing protected information from neural activations [De-Arteaga et al., 2019, Avitan et al., 2025]. This paper focuses on the removal side of that problem: given fixed representations, the goal is to reduce recoverable target-concept information while preserving unrelated information.

Recent studies have primarily examined linear concept erasure [Ravfogel et al., 2020, Belrose et al., 2023], where the goal is to remove linearly decodable concept information from a representation. But nonlinear concept erasure remains unresolved: kernelized adversarial erasure can block a chosen nonlinear adversary, yet protection need not transfer to other nonlinear adversaries [Ravfogel et al., 2022]. This leaves nonlinear concept erasure as an important open problem for representation-level bias mitigation and related interventions.

We approach this vulnerability through a local manifold view of representation space. Hidden states live in a high-dimensional space, but the representations used by natural inputs need not fill it; instead, they may concentrate near a much smaller local support. Because unconstrained edits have the freedom to push representations off this natural support, they can distort independent features while leaving the target concept recoverable.

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This suggests a different design principle for nonlinear erasure: rather than editing along the full gradient, we estimate a local tangent approximation around each sample and retain only the component of the update supported by that local geometry. In practice, we approximate the tangent space via the SVD of a local neighborhood-secant matrix. This is a first-order approximation rather than a theorem about hidden-state geometry, but it yields a concrete and testable erasure method.

Our contributions are twofold:

1. **A manifold-motivated formulation of nonlinear concept erasure.** We frame nonlinear erasure through a local manifold hypothesis: the relevant question is not only whether representations are low-rank, but whether editing should respect the local support rather than act directly in ambient space.
2. **A tangent-constrained nonlinear erasure method.** In Section 4, we introduce TanGCE, which operationalizes this view by projecting ambient erasure updates onto locally estimated tangent spaces so that interventions follow the first-order geometry supported by nearby representations.

We therefore evaluate with a recoverability-oriented protocol: after erasure, we retrain a fresh nonlinear probe and iterate the intervention to remove residual signal. Across evaluations spanning 13 language models across five families and a CLIP-based CelebA benchmark, TanGCE is the stronger nonlinear remover than its ambient ablation and the earlier methods we compare against. The geometry results support the manifold motivation in a bounded form: representations show local low-rank structure, but ambient edits can still leave the natural support. Section 2 positions the method against prior erasure work, Section 4 gives the method and modeling view, Section 5 gives the experimental setup, and Section 6 evaluates both erasure performance and the geometry motivating local tangent constraints.

2 Related Work

Concept erasure methods differ mainly in the function class against which they erase and in the geometry they assume. Linear methods such as INLP [Ravfogel et al., 2020] and LEACE [Belrose et al., 2023] target linearly decodable concept information. INLP removes successively exposed linear directions by iterative nullspace projection, while LEACE gives the strongest closed-form linear baseline in our comparisons. These methods are essential reference points, but they do not directly address the case where a fresh nonlinear probe can still recover the concept after linear erasure.

That recoverability concern is central to our framing. Elazar and Goldberg [Elazar and Goldberg, 2018] showed that apparent success against one learned remover or classifier does not imply that the protected information is gone for a newly trained downstream model. Amnesic probing [Elazar et al., 2021] and related intervention-based analyses adopt the same conservative lesson: the meaningful question is what remains recoverable after the intervention, not only what the original scorer no longer captures.

Among nonlinear erasure methods, IGBP [Iskander et al., 2023] is the closest prior baseline because it iteratively trains a nonlinear probe and retracts representations toward the current decision boundary. Our paper differs in two ways. First, AmbGCE is introduced only as the direct ambient-space ablation that keeps the iterative nonlinear scorer loop but removes the manifold constraint. Second, TanGCE is the paper's main method: it projects the erasure update onto a locally estimated tangent space, making the geometric hypothesis itself the main point of comparison.

3 Problem Setup and Geometric Motivation

We first define the representation-editing problem, then introduce the geometric language used throughout the paper. Our notation follows the manifold viewpoint in Tu's introduction to smooth manifolds [Tu, 2011]: we use local coordinate maps, tangent spaces, and differentials only as local approximations, not as theorem-level guarantees about neural representations.

Language model and encoder. Let Σ^* denote the set of all strings over a vocabulary Σ. A language model p defines a distribution over Σ^*. We write $\mathrm{enc}_\ell : \Sigma^* \to \mathbb{R}^d$ for the hidden representation at

layer ℓ, where d is the hidden dimension. For a fixed layer ℓ (omitted when clear from context), the representation of string s is $\mathbf{x} = \mathrm{enc}_\ell(s) \in \mathbb{R}^d$.

Concepts and data. A binary concept is defined by a labeling function $\phi : \Sigma^* \to \mathcal{C}$, where $\mathcal{C} = \{0, 1\}$. Each sample i has representation $\mathbf{x}_i \in \mathbb{R}^d$ and concept label $y_i = \phi(s_i) \in \mathcal{C}$, with an optional control label z_i (e.g., profession for gender erasure). We write $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ for the dataset and $\mathcal{D}_c = \{i : y_i = c\}$ for the class-c subset, with $N_c = |\mathcal{D}_c|$.

The class-conditional representation is the random variable $X_c \sim \mathrm{enc}_\ell(s) \mid \phi(s) = c$, i.e., the distribution of hidden states for strings labeled c.

Concept scorer. A concept scorer $f_\theta : \mathbb{R}^d \to \mathbb{R}$ is a differentiable nonlinear function trained to predict y from $\mathbf{x}$. In the experiments we instantiate it as a two-layer MLP trained with binary cross-entropy, but the geometric construction below depends only on the scorer being differentiable. Its per-sample gradient with respect to the input is $\nabla_{\mathbf{x}} f_\theta(\mathbf{x}_i) \in \mathbb{R}^d$, which points in the direction of steepest increase in concept probability at $\mathbf{x}_i$.

Representation manifold. Each layer representation $\mathbf{x} = \mathrm{enc}_\ell(s)$ is formally a point in the ambient hidden space $\mathbb{R}^d$, where d is the full hidden-state dimension. We adopt a local manifold hypothesis: the representations that occur for natural inputs lie near a smooth embedded submanifold $\mathcal{M} \subset \mathbb{R}^d$, or more cautiously, near a locally low-dimensional support. This is a modeling assumption rather than a theorem about neural representations. Its operational consequence is local: every smooth manifold is, by definition, *locally Euclidean* — in a neighborhood of any point $\mathbf{x} \in \mathcal{M}$ there exists a coordinate chart that maps onto an open subset of $\mathbb{R}^m$ [Tu, 2011] — so it is meaningful to approximate the neighborhood of $\mathcal{M}$ around $\mathbf{x}$ by its tangent space $T_{\mathbf{x}}\mathcal{M}$ and to apply ordinary linear-algebraic operations (SVD, projection, anisotropic scaling) inside it. In the method below, this tangent space is not observed directly; it is estimated from nearby representations by local SVD on the secant matrix.

Now fix a point $\mathbf{x} \in \mathcal{M}$. Let $U \subset \mathcal{M}$ be a small local neighborhood around $\mathbf{x}$, and let

$$\varphi : U \to \Omega \subset \mathbb{R}^m \tag{1}$$

be a local coordinate map. Here Ω is an open set in the coordinate space $\mathbb{R}^m$. Intuitively, φ assigns m local coordinates to nearby points on the manifold. We choose the coordinates so that $\varphi(\mathbf{x}) = 0$, meaning that the point $\mathbf{x}$ is represented by the origin in local coordinates. The inverse map $\varphi^{-1} : \Omega \to U \subset \mathcal{M} \subset \mathbb{R}^d$ is the corresponding local parametrization back into the ambient hidden space.

The tangent space at $\mathbf{x}$ is the best local linear approximation to the manifold near $\mathbf{x}$. Formally, it is the m-dimensional linear space

$$T_{\mathbf{x}}\mathcal{M} = d(\varphi^{-1})_0(\mathbb{R}^m) \subset \mathbb{R}^d, \tag{2}$$

where $d(\varphi^{-1})_0$ is the differential of the local parametrization φ^{-1} at the origin. In ordinary ML language, this differential can be read as the Jacobian or best local linear approximation of φ^{-1} at 0. It maps an infinitesimal coordinate displacement in $\mathbb{R}^m$ to the corresponding first-order displacement in the ambient space $\mathbb{R}^d$. Because $\mathcal{M}$ is embedded in $\mathbb{R}^d$, we can treat $T_{\mathbf{x}}\mathcal{M}$ as an m-dimensional linear subspace of the ambient hidden space.

Ambient versus tangent directions. For any *ambient vector* $\mathbf{g} \in \mathbb{R}^d$ attached to the point $\mathbf{x}$, the local geometric picture suggests a decomposition

$$\mathbf{g} = \mathbf{g}^\top + \mathbf{g}^\perp, \qquad \mathbf{g}^\top \in T_{\mathbf{x}}\mathcal{M}, \quad \mathbf{g}^\perp \perp T_{\mathbf{x}}\mathcal{M}. \tag{3}$$

Here $\mathbf{g}^\top$ is the tangent component, namely the part of $\mathbf{g}$ that lies in the estimated tangent space, and $\mathbf{g}^\perp$ is the normal component, namely the part orthogonal to it. In this local approximation, the tangent component changes the representation along directions spanned by nearby samples in the training set, while the normal component is orthogonal to that local linear subspace. This is the geometric distinction used later by TanGCE: ambient gradient updates may contain both parts, but tangent-projected updates keep only the component that remains aligned with the estimated local geometry. We do not claim that the normal component is causally responsible for off-manifold displacement, only that the tangent projection is a practical way to attenuate directions unsupported by the local sample structure.

Algorithm 1 TanGCE at a glance. Phase 1 is two closed-form, one-shot projections (LEACE then a rank-2 covariance-asymmetry projection); Phase 2 is an iterative tangent-constrained step on the residual. Mathematical details in §4.1–4.6.

Require: Representations $\mathbf{X}$, concept labels $\mathbf{y}$, #rounds T, trust region ε, anisotropy α, scorer hidden h.

Phase 1 – closed-form moment preprocessing (Eqs. 4–6):

  1a. Linear: $\mathbf{X} \leftarrow \mathrm{LEACE}_{\mathbf{y}}(\mathbf{X})$ (rank-1 mean projection)

  1b. Covariance: $\mathbf{X} \leftarrow \mathbf{X} - \mathbf{D}\mathbf{D}^\top\mathbf{X}$, with $\mathbf{D}$ = top-2 eigvecs of $\Delta\mathbf{\Sigma}$ (rank-2 covariance projection)

Phase 2 – iterative tangent step (Eqs. 10–19):

for $t = 1, \ldots, T$ **do**

  Train a fresh 2-layer MLP scorer f_t (hidden h) on the current $(\mathbf{X}, \mathbf{y})$.

  For each row $\mathbf{x}_i$: take a tangent-projected, σ-preconditioned trust-region step

$$\mathbf{x}_i \leftarrow \mathbf{x}_i - \lambda_i \cdot \underbrace{\mathbf{C}_i^{\alpha}}_{\substack{\text{anisotropy} \\ (\sigma_{i,k}^{2\alpha} \text{ per axis})}} \cdot \underbrace{\mathbf{P}_{T_i}\nabla f_t(\mathbf{x}_i)}_{\text{tangent-projected gradient}}, \quad \text{with } \|\mathbf{x}_i \text{ step}\|_2 \le \varepsilon \cdot s_i.$$

end for

return edited $\mathbf{X}$.

Recoverability after erasure. Throughout the paper, concept removal is judged only after training a *fresh* probe on the edited representations. This follows the lesson of Elazar and Goldberg [2018]: erasure succeeds only if a newly trained classifier still fails to recover the concept from $\tilde{\mathbf{x}}$.

Conventions. Index i ranges over samples. $\|\cdot\|$ denotes the Euclidean norm unless stated otherwise.

4 Method: TanGCE

We now introduce *Tangent Gradient-based Concept Erasure* (TanGCE), the method presented in this paper. The full pipeline has two phases. **Phase 1** is two closed-form, one-shot moment-matching projections that exhaust all first- and second-moment concept signal in the representation. **Phase 2** is the iterative tangent-constrained loop that erases the residual nonlinear concept structure left after Phase 1. The starting point of Phase 2 is a local manifold approximation of representation space: if the neighborhood of a representation $\mathbf{x}$ is well approximated by a low-dimensional manifold $\mathcal{M}$, then the relevant first-order directions are those in the tangent space $T_{\mathbf{x}}\mathcal{M}$ rather than arbitrary ambient directions in $\mathbb{R}^d$. We do not observe this tangent space directly. Instead, we approximate it via the local SVD of the neighborhood secant matrix and use that first-order estimate to constrain the erasure update. Each iteration of Phase 2 therefore follows the same four-step pattern: train a fresh concept scorer, estimate a local tangent basis, project the scorer gradient into that tangent space, and apply the resulting constrained update.

4.1 Phase 1: Closed-form moment preprocessing

Before iteration begins, we apply two closed-form, one-shot moment-matching projections that exhaust all linear (1st-moment) and rank-2 quadratic (2nd-moment) concept signal in the representation. The iterative tangent loop then operates on the residual nonlinear structure that no closed-form linear projection can reach.

LEACE (1st moment). We fit a LEACE eraser [Belrose et al., 2023] on $(\mathbf{X}_{\text{train}}, \mathbf{y})$ and apply it to all three splits. LEACE removes the rank-1 cross-covariance $\mathbf{\Sigma}_{xz}$: after this step, no *linear* classifier can recover the concept above majority-class accuracy on training data, with provably minimum representation damage among affine projections enforcing this invariant. In our setting LEACE removes one rank from a representation of dimension $d \in \{2048, 3072, 3584\}$ at negligible utility cost.

Covariance-asymmetry projection (2nd moment). LEACE matches class means but leaves the within-class covariance asymmetry

$$\Delta\Sigma \;=\; \Sigma_+ - \Sigma_- \tag{4}$$

unchanged, where $\Sigma_\pm$ are the per-class within-class covariances of the LEACE-projected features. A 2-layer MLP probe can still recover concept information from variance asymmetry along axes where Σ_+ and Σ_- differ, even when the class means are equal — squared features expose this signal that is invisible to any linear classifier. We therefore append a rank-2 projection: take the top-2 eigenvectors of $\Delta\Sigma$ ranked by $|\lambda|$, orthonormalize with a residual mean direction via QR,

$$\mathbf{D} \;=\; \mathrm{QR}\Big(\big[\,\hat{\mathbf{d}}_{\text{mean}} \;\big|\; \mathbf{e}_1, \mathbf{e}_2\,\big]\Big) \;\in\; \mathbb{R}^{d\times 3}, \tag{5}$$

and project this rank-3 subspace out of every split:

$$\tilde{\mathbf{x}} \;=\; \mathbf{x} \;-\; \mathbf{D}\mathbf{D}^\top\mathbf{x}. \tag{6}$$

Because LEACE has already aligned the class means, $\hat{\mathbf{d}}_{\text{mean}} \approx \mathbf{0}$ and the QR step keeps only the two eigenvector directions in practice, so the effective preprocessing rank removed is two. This step is a special case of the k-LEACE-style higher-moment erasure framework of Singh et al. [2024]; we use the rank-2 instance because second-order eigenvectors are stable to estimate from $n \approx 7000$ training examples while higher-order tensors are not.

Why two stages, both closed-form. Each stage removes a structurally distinct part of the concept signal that the other cannot reach: LEACE eliminates the mean-shift any linear classifier exploits; the covariance projection eliminates the variance asymmetry a quadratic classifier exploits. Both stages have closed-form solutions on full training data, making them more accurate than per-step or kernelized estimates, and each costs $\mathcal{O}(d^2)$ time and a single rank-1-or-2 projection at inference. They remove rank ≤ 3 dimensions out of $d \in \{2048, 3072, 3584\}$ — a negligible fraction of the representation budget — while exposing the residual nonlinear concept structure that the iterative loop in Phase 2 is designed to erase.

4.2 Local tangent estimation

For each query point $\mathbf{x}_i$, we estimate the tangent space $T_{\mathbf{x}_i}\mathcal{M}$ via the SVD of a local secant matrix. Let $\mathcal{N}_k(\mathbf{x}_i) = \{\mathbf{x}_{j_1}, \ldots, \mathbf{x}_{j_k}\}$ denote the k nearest neighbors of $\mathbf{x}_i$ in the training set, excluding $\mathbf{x}_i$ itself. We stack the centered secant vectors $\mathbf{s}_{j_m} = \mathbf{x}_{j_m} - \mathbf{x}_i$ row-wise into the secant matrix $\mathbf{S}_i \in \mathbb{R}^{k\times d}$ and compute its SVD,

$$\mathbf{S}_i \;=\; \mathbf{U}_i \,\mathrm{diag}(\sigma_{i,1}, \ldots, \sigma_{i,k})\, \mathbf{V}_i^\top, \tag{7}$$

which yields the right singular vectors $\mathbf{V}_i \in \mathbb{R}^{d\times k}$ (the directions in which neighbors extend) and the singular values $\sigma_{i,k}$ (how far they extend along each direction). The implied local covariance — the second moment of the secants in the ambient space — is then $\mathbf{C}_i = \frac{1}{k}\mathbf{S}_i^\top\mathbf{S}_i = \frac{1}{k}\mathbf{V}_i\,\mathrm{diag}(\sigma_{i,k}^2)\,\mathbf{V}_i^\top$:

$$\mathbf{C}_i = \frac{1}{k}\sum_{m=1}^{k} \mathbf{s}_{j_m}\mathbf{s}_{j_m}^\top \in \mathbb{R}^{d\times d}. \tag{8}$$

Truncating to the top r right singular vectors gives the tangent basis

$$\mathbf{B}_i = [\mathbf{v}_i^{(1)}, \ldots, \mathbf{v}_i^{(r)}] \in \mathbb{R}^{d\times r}, \tag{9}$$

where the per-panel intrinsic dimension r is estimated by the *TwoNN* algorithm of Facco et al. [2017], which exploits the ratio of the first two nearest-neighbor distances around each point to give a maximum-likelihood estimate of the local manifold dimension without any assumption on the neighborhood scale. We apply TwoNN once per panel and floor the result at $r \geq 8$ to keep the tangent basis well-conditioned even on panels where the estimator returns very small values; the floor never binds in practice on the language-model panels we evaluate. The resulting orthogonal projector onto the estimated tangent space is

$$\mathbf{P}_{T_i} = \mathbf{B}_i\mathbf{B}_i^\top. \tag{10}$$

This is the operational approximation used in TanGCE: the leading right singular vectors of the local secant matrix serve as an empirical linear basis for $T_{\mathbf{x}_i}\mathcal{M}$. We treat the SVD basis as a practical first-order stand-in for a tangent space, not as a theorem-level guarantee that the representations form a smooth manifold.

4.3 Tangent-constrained update

To define the tangent-constrained update, we start from the ambient unit gradient direction

$$\mathbf{u}_i = \frac{\nabla_{\mathbf{x}} f_\theta(\mathbf{x}_i)}{\|\nabla_{\mathbf{x}} f_\theta(\mathbf{x}_i)\|}. \tag{11}$$

TanGCE then defines the tangent-constrained direction by projecting onto the local tangent space:

$$\mathbf{u}_i^{\tan} = \mathbf{P}_{T_i}\,\mathbf{u}_i = \mathbf{B}_i(\mathbf{B}_i^\top \mathbf{u}_i). \tag{12}$$

The projected direction is *not* re-normalized to unit length. Its norm $\|\mathbf{u}_i^{\tan}\| \leq 1$ acts as an overlap weight: when the ambient concept gradient has little support in the estimated tangent space, the update is naturally attenuated.

TanGCE then applies the update

$$\tilde{\mathbf{x}}_i = \mathbf{x}_i - \lambda\,(\mathbf{x}_i^\top \mathbf{u}_i^{\tan})\,\mathbf{u}_i^{\tan}. \tag{13}$$

where λ is a step-size scalar; in §4.5 it becomes a per-sample λ_i chosen by a closed-form trust-region rule. The update in Eq. 13 is a heuristic design choice, not the solution to an explicit optimization problem. It projects the representation onto the estimated tangent direction and subtracts a scaled multiple, analogous to a one-step descent along the constrained direction.

The effective step size is $\lambda\|\mathbf{u}_i^{\tan}\|^2$. It is larger when the gradient has more support in the estimated tangent space and smaller when it has less, but this does not guarantee monotonic improvement in either leakage or utility.

This construction is the core difference between TanGCE and its ambient counterpart. AmbGCE applies the update in the full hidden space, whereas TanGCE restricts the update to directions supported by the local geometry estimated in the model's native coordinate system. We do not claim invariance under coordinate changes; the tangent basis is computed in the LayerNorm-induced geometry of the given representations.

Equation 13 treats every direction within the tangent space symmetrically. We retain it here as the conceptual entry point and as the $\alpha = 0$ ablation, but the canonical TanGCE update used throughout the paper is the σ-preconditioned generalization derived in §4.6 (Eq. 19), which respects the anisotropy of the local manifold by reweighting each tangent direction with its associated singular value.

4.4 Why tangent projection can help at larger λ

AmbGCE edits $\mathbf{x}_i$ in the full ambient space, where the gradient may contain components that are poorly aligned with the local representation support. Under the local manifold view, those components are the ones most likely to move the representation away from the neighborhood geometry captured by natural data. TanGCE removes them at first order by replacing $\mathbf{u}_i$ with its projection $\mathbf{u}_i^{\tan}$ into the estimated tangent space. This motivates the hypothesis that tangent-constrained updates can remain effective at larger step sizes. We treat that statement as empirical rather than guaranteed, and evaluate it directly in the results section.

4.5 Adaptive per-sample step size: closed-form λ_i

A single global λ over-edits some samples while under-editing others. Confident samples (large gradient norm and well-aligned tangent) move proportionally more for any fixed scalar; uncertain samples (small or off-manifold gradient) are dragged by the same scalar in noise-dominated directions. We address this with a per-sample step size λ_i that respects the *local trust region* of the manifold approximation.

Trust region. Recall that $\mathbf{B}_i$ is the local SVD basis at $\mathbf{x}_i$ (the top-r right singular vectors of the secant matrix, Eq. 9) and let s_i be a per-sample scale on which the linear tangent approximation is reliable. We require that no sample moves further than a fraction ε of this local scale:

$$\|\tilde{\mathbf{x}}_i - \mathbf{x}_i\|_2 \;\leq\; \varepsilon \cdot s_i. \tag{14}$$

Here $\varepsilon \in (0, \infty)$ is a single global trust-region scale, dimensionless and shared across panels. Smaller ε yields surgical edits inside each local manifold patch; larger ε admits steps that may leave the patch. We use $\varepsilon = 0.1$ throughout (an order of magnitude smaller than the canonical $\varepsilon = 0.3$ used in our earlier ablations); see § 6 for the empirical effect of this choice on tight-budget surgicality. The simplest choice is $s_i = r_i$, the mean distance to the k nearest neighbors — the radial neighborhood scale. We use a tighter, anisotropy-aware scale by default; see Eq. 17 below.

Closed-form solution. Substituting Eq. 13 into Eq. 14, the constraint becomes

$$\lambda_i \cdot |\mathbf{x}_i^\top \mathbf{u}_i^{\tan}| \cdot \|\mathbf{u}_i^{\tan}\| \;\leq\; \varepsilon \cdot s_i. \tag{15}$$

Let $\mathbf{v}_i = (\mathbf{x}_i^\top \mathbf{u}_i^{\tan})\, \mathbf{u}_i^{\tan}$ be the per-sample edit displacement at λ=1. The largest λ_i satisfying Eq. 15 is

$$\boxed{\lambda_i \;=\; \min\!\left(\lambda_{\max},\; \frac{\varepsilon \cdot s_i}{\|\mathbf{v}_i\|}\right)} \tag{16}$$

where $\lambda_{\max}$ is a safety cap that prevents runaway updates when $\|\mathbf{v}_i\| \to 0$ (gradient nearly orthogonal to the tangent space). In practice we set $\lambda_{\max} = 64$, large enough that it binds only for samples whose gradient has near-zero tangent component.

Anisotropy-aware local scale. Setting $s_i = r_i$ (mean k-NN distance) is the simplest valid choice, but treats every direction equally. The local manifold is generally an *ellipsoid*, not a sphere: along the dominant right singular vectors of the secant matrix, neighbors extend much further than along thin (low-σ) directions. We therefore prefer the *whitened-tangent neighborhood radius*

$$s_i \;=\; R_i^{\text{white}} \cdot s_i^{\text{maha}}, \qquad R_i^{\text{white}} = \text{median}_{j \in \mathcal{N}_k(i)}\, \|\mathbf{B}_i^\top (\mathbf{x}_j - \mathbf{x}_i) \,/\, \boldsymbol{\sigma}_i\|_2, \tag{17}$$

where $\boldsymbol{\sigma}_i = (\sigma_{i,1}, \ldots, \sigma_{i,r})$ is the local singular spectrum and division is component-wise. The factor R_i^{white} measures how far neighbors typically reach in standardized tangent coordinates; the factor $s_i^{\text{maha}} = 1/\|\boldsymbol{\alpha}_i/(\boldsymbol{\sigma}_i + \delta)\|_2$ rescales the per-sample step direction by its harmonic-σ tangent length, where $\boldsymbol{\alpha}_i$ is the unit step direction in tangent coordinates. We refer to this scaling as *fractional Mahalanobis* (FMaHa): under it, ε is interpretable as a fraction of the local whitened-tangent neighborhood. Setting $s_i = r_i$ recovers the original Euclidean radial scale and is reported as an ablation.

Why this matters. Eq. 16 is a genuine *trust region* per sample: confident samples move by exactly $\varepsilon \cdot r_i$ along their tangent; uncertain samples (small $\|\mathbf{v}_i\|$) get a larger nominal λ_i but the actual movement $\lambda_i \cdot \|\mathbf{v}_i\|$ remains bounded by $\varepsilon \cdot r_i$ (or by the safety cap, in which case movement is small because $\|\mathbf{v}_i\|$ is small). This per-sample adaptation removes a hyperparameter (one global λ) and replaces it with a single dimensionless surgicality knob ε.

The closed-form expression in Eq. 16 requires no bisection or auxiliary optimization: each iteration computes $\mathbf{v}_i$ once via a forward pass through the scorer, s_i once via the cached k-NN distances and the local SVD spectrum, and λ_i in $\mathcal{O}(1)$ per row. This makes the per-sample trust region essentially free relative to the cost of refitting the scorer and tangent.

4.6 Anisotropic update via the local σ-spectrum

The closed-form λ_i above controls the *magnitude* of each sample's step. Its *direction* is still isotropic in the tangent basis: every tangent component of $\mathbf{v}_i$ is rescaled by the same scalar λ_i. But the local SVD already tells us that the manifold is anisotropic — squared singular values $\sigma_{i,k}^2$ at sample i differ by orders of magnitude across k. Treating all tangent directions as equally safe to move along is exactly the assumption that the neighborhood is a sphere; in practice it is much closer to an ellipsoid stretched along the dominant right singular vectors and pinched in the rank-truncation tail. An isotropic step over-edits thin directions where the linear approximation is most fragile, and under-edits fat directions where the manifold easily accommodates further displacement.

Singular-vector decomposition. Decompose the per-sample tangent displacement $\mathbf{v}_i$ in the local SVD basis:

$$\mathbf{v}_i \;=\; \sum_{k=1}^{r} a_{i,k}\, \mathbf{v}_i^{(k)}, \qquad a_{i,k} = \langle \mathbf{v}_i, \mathbf{v}_i^{(k)} \rangle, \tag{18}$$

where $\{\mathbf{v}_i^{(k)}\}_{k=1}^r$ are the columns of $\mathbf{B}_i$ (the leading right singular vectors of the local secant matrix, Eq. 7) with associated singular values $\sigma_{i,k}$. The implied local covariance is $\mathbf{C}_i = \mathbf{B}_i \operatorname{diag}(\sigma_{i,1}^2, \ldots, \sigma_{i,r}^2)\, \mathbf{B}_i^\top$, with eigenvalues $\sigma_{i,k}^2$.

Anisotropic update. We replace the isotropic step in Eq. 13 by a covariance-preconditioned step that respects the local ellipsoid. Define

$$\boxed{\tilde{\mathbf{x}}_i \;=\; \mathbf{x}_i - \lambda_i\, \mathbf{C}_i^{\alpha}\, \mathbf{v}_i \;=\; \mathbf{x}_i - \lambda_i \sum_{k=1}^{r} \sigma_{i,k}^{2\alpha}\, a_{i,k}\, \mathbf{v}_i^{(k)}} \tag{19}$$

where $\alpha \geq 0$ is an anisotropy exponent. The effect is direct: each tangent coefficient $a_{i,k}$ is reweighted by $\sigma_{i,k}^{2\alpha}$, so the update moves more along high-variance principal directions (where the manifold supports larger displacements) and less along thin directions (where the linear approximation is fragile).

Three regimes. Three values of α have natural interpretations:

- $\alpha = 0$: $\mathbf{C}_i^0 = \mathbf{I}$, the update reduces to Eq. 13. Reported as an ablation.
- $\alpha = \frac{1}{2}$: a *mild* anisotropy. Each direction is reweighted by $\sigma_{i,k}$ rather than by $\sigma_{i,k}^2$, preserving information from low-variance directions while still respecting the dominant geometry. **This is the default in this paper.**
- $\alpha = 1$: full Mahalanobis preconditioning. The update direction $\mathbf{C}_i\mathbf{v}_i$ is the first-order optimum of $\min_{\boldsymbol{\delta}} \langle \mathbf{g}_i, \boldsymbol{\delta} \rangle$ subject to the Mahalanobis trust region $\boldsymbol{\delta}^\top \mathbf{C}_i^{-1} \boldsymbol{\delta} \leq \varepsilon^2$. Reported as an ablation; risks suppressing low-variance signal that may carry residual concept information.

Relation to Mahalanobis whitening (and what $\alpha = \frac{1}{2}$ is *not*). A reader familiar with Mahalanobis whitening might wonder whether $\alpha = \frac{1}{2}$ is "the whitening" we are doing. It is not. Mahalanobis whitening transforms the data so the covariance becomes identity, $\mathbf{x}^{\text{white}} = \mathbf{C}^{-1/2}\mathbf{x}$, and corresponds to the *negative* half-power of $\mathbf{C}$ (i.e. $\alpha = -\frac{1}{2}$ in our notation). Whitening makes every direction equal-weighted, which would actively dilute the concept-specific signal we want to suppress and force every tangent axis to receive equal step magnitude.

We use $\mathbf{C}^{+1/2}$ (positive half-power), which does the opposite: it reweights the tangent step *toward* the high-$\sigma_{i,k}$ principal axes — exactly the directions where the local manifold has the most support and where the linear approximation is most reliable — and away from low-σ thin tail directions where the linear approximation is fragile and where motion most easily leaves the manifold. The three values $\alpha \in \{0, \frac{1}{2}, 1\}$ thus form a one-parameter family interpolating between (i) treating every tangent direction as equally trustworthy ($\alpha = 0$, isotropic), (ii) respecting the local ellipsoid mildly with $\sigma_{i,k}$ scaling ($\alpha = \frac{1}{2}$, our default), and (iii) the natural-gradient / Mahalanobis trust-region optimum that maximally suppresses thin directions ($\alpha = 1$). Whitening ($\alpha = -\frac{1}{2}$) sits outside this useful range for our problem because it has the opposite sign of the geometric correction we want; we therefore do not include it.

Trust region preserved. The trust-region magnitude in Eq. 16 is enforced after preconditioning, against the transformed step $\mathbf{C}_i^{\alpha}\mathbf{v}_i$ (i.e. we substitute $\|\mathbf{C}_i^{\alpha}\mathbf{v}_i\|$ for $\|\mathbf{v}_i\|$ in the denominator). The actual displacement $\|\tilde{\mathbf{x}}_i - \mathbf{x}_i\|_2$ therefore remains bounded by $\varepsilon \cdot s_i$, exactly as in the isotropic case. Anisotropy reshapes the *direction* of the step, not its budget.

Implementation. Eq. 19 reuses the SVD already computed for the tangent basis; the per-sample singular spectrum $(\sigma_{i,k})_k$ is cached during the local SVD. Concretely, the step decomposition (Eq. 18), eigenvalue reweighting, and reprojection to the ambient space are three batched einsums over the (N, d, r) basis tensor. No additional SVDs and no per-sample loops are required. TanGCE therefore inherits the same per-iteration cost as the isotropic update, modulo a single extra $\mathcal{O}(Ndr)$ matrix multiplication.

4.7 Iterative TanGCE and relation to baselines

We implement TanGCE using an iterative protocol with probe recycling. Each round begins by training a fresh concept scorer on the current representations, computing per-sample gradients, and applying the tangent-constrained update in Eq. 13. This turns the method into an iterative nonlinear eraser: every new scorer is trained specifically to recover whatever target information survived the previous round. The tangent estimator is also refit in every round to track the evolving local geometry.

The rationale follows Elazar and Goldberg [Elazar and Goldberg, 2018]: success against a single trained classifier is not evidence that the concept has been exhausted. We therefore evaluate every erasure method against a retrained probe after each full erasure procedure, and the erasure itself iterates to remove residual recoverable signal. This nonlinear fresh-scorer loop is the conceptual analogue of iterative linear removal in INLP [Ravfogel et al., 2020], but with a newly trained nonlinear scorer at each round.

This setup also clarifies the relation between TanGCE and the baselines. IGBP [Iskander et al., 2023] is the closest prior nonlinear baseline, performing a one-shot correction to the current decision boundary in each round. AmbGCE is the direct ablation of TanGCE: it keeps the same fresh-scorer loop but applies the update in the full ambient space. TanGCE is the full method, adding the tangent-space restriction.

Two diagnostics monitor tangent quality throughout the iterative process. *Tangent Gradient Capture* (TGC) measures the fraction of the leakage gradient lying within the estimated tangent space. *Tangent Neighbor Residual* (TNR) measures the fraction of local secant variation not explained by the tangent basis. High TGC and low TNR indicate that the local linear approximation is informative enough for the tangent-constrained update to be meaningful.

5 Experimental Setup

We evaluate each erasure method by training all probes from scratch on the intervened representations, so that success cannot be attributed to leakage from the erasure scorer itself.

5.1 Evaluation Protocol

Fresh-probe evaluation. The precision metric is intentionally strict: after every erasure method finishes, we train a *new* probe on the intervened representations. If a freshly trained probe can still recover the concept, then the concept was not actually removed in the sense relevant for downstream use.

Shared validation protocol. Most gradient-based methods share a common erasure strength parameter λ. For AmbGCE and TanGCE, λ scales the ambient or tangent-projected update. Because TanGCE's tangent projection attenuates the effective step size by $\|\mathbf{u}_i^{\tan}\|^2 \leq 1$, a fixed λ across methods would systematically under-erase for the tangent-constrained update. Rather than manually choosing different values, we adopt validation-based hyperparameter selection: all λ-based methods search the same grid and select the λ^* that minimizes validation nonlinear probe accuracy. The layer-8 single-model comparison (Appendix Table 8) and vision benchmarks use $\lambda \in \{0.5, 1.0, 2.0, 4.0, 8.0\}$; the multi-model generalization experiments use the extended grid $\lambda \in \{0.25, 0.5, 1.0, 2.0, 4.0, 8.0, 16.0, 32.0\}$, which we found necessary for larger models and deeper layers where optimal λ^* can exceed 8.0. IGBP has no explicit λ parameter—its step size is determined by each sample's distance to the decision boundary—so it runs until convergence.

Training protocol. All MLP probes—the shared *concept scorer* used as the erasure direction source, the *validation probe* for λ HPO, and the *evaluation probe* retrained on intervened representations—share the same architecture: two-layer MLP with hidden dimension 256, AdamW (lr $= 10^{-3}$, weight decay 10^{-4}), batch size 512. For language model experiments, all three probes use 1500 gradient steps. For CelebA, the concept scorer uses 1000 steps and the HPO and evaluation probes use 1200 steps; the shorter budget suffices for the lower-dimensional CLIP embeddings. Using identical probe budgets for HPO and evaluation within each modality ensures that the lambda selected on validation is directly comparable to the final test metric, with no ranking distortion from mismatched model capacity. The tangent estimator uses k=50 nearest neighbors for local PCA; the retained tangent rank

per sample is determined by a retained-variance threshold applied to the local eigenspectrum. IGBP runs for up to 35 rounds; each round trains its probe for 10 epochs with AdamW lr $= 2\times10^{-4}$, batch size 256, and stops early when validation accuracy falls to majority-class level. Linear baselines (LEACE, INLP, Linear Projection) use ℓ_2-regularized logistic regression with balanced class weights. The control-task predictor is also an ℓ_2 logistic regression trained on clean representations and evaluated on the intervened test set without retraining.

Domain-specific controls. For sycophancy we use answer matching as the control concept; for Bias in Bios we use profession; for safety we use helpfulness; and for CelebA we use selected low-correlation facial attributes (5 per concept). This lets the evaluation distinguish targeted concept removal from broad representational damage.

Vision benchmarks. For cross-modal evaluation we use CelebA [Liu et al., 2015], which provides 202,599 celebrity images with 40 binary facial attributes; we use the official train/val/test split. We extract pooled CLIP ViT-B/32 image embeddings (768-dim) and apply all erasure methods to the resulting representation matrix.

5.2 Metrics

Erasure precision. Let h^{NL} be a nonlinear probe and h^{L} a linear probe trained on $\tilde{\mathbf{x}}_i$ to predict the target concept y_i. We report their test accuracies on the intervened representations:

$$\mathrm{Prec}^{\mathrm{NL}}_{\mathrm{erase}} = \mathrm{Acc}(h^{\mathrm{NL}}(\tilde{\mathbf{X}}), y), \qquad \mathrm{Prec}^{\mathrm{L}}_{\mathrm{erase}} = \mathrm{Acc}(h^{\mathrm{L}}(\tilde{\mathbf{X}}), y). \tag{20}$$

Lower values indicate stronger erasure. Because the target concepts in this paper are not assumed to be linearly encoded, the retrained nonlinear probe is the primary precision metric and the linear probe is a secondary diagnostic. For interpretation, the majority-class predictor provides a trivial lower-bound reference for what it means to erase by collapsing the target signal entirely rather than preserving useful structure.

Surgicality on control concepts. Let q be a predictor for an annotated control label c (e.g., profession for gender erasure, helpfulness for safety, or selected low-correlation attributes in CelebA). We report the change in control performance relative to the clean representations:

$$\Delta_{\mathrm{ctrl}} = \mathrm{Acc}(q(\tilde{\mathbf{X}}), c) - \mathrm{Acc}(q(\mathbf{X}), c). \tag{21}$$

Smaller $|\Delta_{\mathrm{ctrl}}|$ means the erasure is more surgical. In CelebA, where each target attribute is paired with multiple controls, we report the mean Δ_{ctrl} across the selected controls.

Language-model utility via cache-replay perplexity. Probe-based recoverability and control-task surgicality both bypass the language-model head: they read off the intervened representation $\tilde{\mathbf{x}}$ directly. To check that erasure does not silently break the LM's downstream behaviour, we additionally measure next-token perplexity when the erased representation is plugged back into the residual stream. Concretely, for each evaluation example we capture the layer-ℓ residual at the labelled token position, compute its erased counterpart $\tilde{\mathbf{x}} = f_{\mathrm{erase}}(\mathbf{x})$ offline, and re-run the forward pass with a hook that replaces the position's residual with $\tilde{\mathbf{x}}$ before continuing through the rest of the model. We report the resulting cross-entropy on the post-replacement suffix tokens together with $\Delta_{\mathrm{PPL}} = \mathrm{PPL}_{\tilde{\mathbf{x}}} - \mathrm{PPL}_{\mathbf{x}}$, with values averaged over three seeds. Smaller Δ_{PPL} means the erasure preserves more of the LM's continuation behaviour on the very inputs it targets.

Together, these metrics ask whether an erasure method removes the target concept while preserving unrelated control concepts and the LM's local continuation behaviour.

5.3 Datasets, Models, and Compared Methods

This subsection completes the experimental setup by describing the datasets, models, and comparison methods used to evaluate TanGCE.

5.3.1 Experimental Domains

We evaluate our methods across four evaluation settings:

1. **Sycophancy (Primary Nonlinear Case):** Our central focus is on removing agreement bias in model responses. We use the Anthropic model-written-evals sycophancy suite [Perez et al., 2023], which comprises three subsets: NLP survey, PhilPapers 2020, and political typology quiz. Each example is a multiple-choice question prefaced by a persona bio that asserts a position, with a designated *sycophantic* answer (the one matching the persona's assertion) and one or more *non-sycophantic* alternatives. Because the dataset does not provide pre-assigned per-example concept labels, we assign them by running the model itself: concept $= 1$ if the model's argmax prediction matches the sycophantic answer, concept $= 0$ otherwise. This labelling strategy identifies exactly which examples activate the sycophancy direction in the model being studied, rather than relying on a fixed human annotation. For the surgicality control we use *answer matching*: whether the model selects the non-sycophantic (factually correct) answer. This adapts the standard helpfulness control from safety datasets to the sycophancy setting, where "helpful" behaviour is operationalized as resisting social pressure and answering correctly.
2. **Safety (Secondary Case Study):** We evaluate the surgicality of erasure on content safety using the PKU-SafeRLHF dataset [Ji et al., 2024], testing the preservation of helpfulness alongside safety leakage.
3. **Bias in Bios (Diagnostic Baseline):** We use the Bias in Bios dataset [De-Arteaga et al., 2019] for gender and profession classification as a replication anchor. This setting provides a linear comparison point to contrast against the nonlinear gains of tangent-constrained erasure.
4. **CelebA Visual Attributes (Cross-Modal Generalization):** We apply the same erasure methods to CLIP ViT-B/32 [Radford et al., 2021] pooled embeddings (768 dimensions) of CelebA [Liu et al., 2015] face images. CelebA provides 40 binary facial attributes, enabling a rich surgicality analysis: when erasing one attribute (e.g., *Male*), the remaining 39 serve as potential controls. We select the 5 least-correlated attributes as controls per concept to ensure that control accuracy changes reflect genuine collateral damage rather than attribute correlation.

5.3.2 Models

Language models. The main seven-method comparison is run on Qwen2.5-0.5B [Qwen Team, 2024] (24 decoder layers, hidden dimension 896) at layer 8. To test whether the tangent-vs.-ambient gap persists across families and scales, we additionally compare TanGCE and AmbGCE on Qwen2.5-1.5B, Qwen2.5-3B, Gemma-2-2B, Gemma-2-9B, Gemma-2-27B, Llama-3.2-1B, Llama-3.2-3B, Mistral-7B-v0.1, Gemma-3-1B, Gemma-3-4B, Gemma-3-12B, and Gemma-3-27B. **For each model we extract residual-stream representations from the layer at 50% of total decoder depth** (e.g., layer 12 of 24 for Qwen2.5-0.5B, layer 16 of 32 for Mistral-7B, layer 24 of 48 for Gemma-3-27B). We pick mid-depth because (i) early layers encode mostly token-level features unrelated to the high-level concepts we erase, (ii) late layers risk degenerate task-specific representations that interact unpredictably with the LM head when re-injected, and (iii) on a held-out validation panel the 50% layer was best or tied for both control utility and concept recoverability across all families we tested. The full Pareto comparison against Obliviator (Section 6.2, Figure 1) reports the 39-panel grid (13 LLM families $\times$ 3 concepts) at this 50%-depth layer.

Vision encoder. The CelebA and FairFace experiments use CLIP ViT-B/32 pooled embeddings (768 dimensions), extracted once and cached. We keep the encoder fixed across the two vision benchmarks so that the comparison isolates dataset effects rather than encoder differences.

5.3.3 Compared Methods

Methods. The core quantitative comparison evaluates TanGCE against eight baselines and ablations, plus a Majority Class reference floor: two random baselines (global and per-sample direction), linear projection, INLP [Ravfogel et al., 2020], LEACE [Belrose et al., 2023], IGBP [Iskander et al., 2023], Obliviator [Shao et al., 2024], and AmbGCE. **Obliviator** is the current state-of-the-art unsupervised concept-erasure baseline: an HSIC-encoder + RFF-EVP iterative projection that removes the empirically estimated nonlinear concept signal without using control labels. IGBP is the closest prior *supervised* nonlinear baseline and serves as the main methodological reference point

Table 1: Comparison of concept erasure methods and reference baselines by family and role in this work.

Method	Family	Role in Paper
Majority Class	Reference	Trivial no-information predictor used as a recoverability floor, not as an erasure method
Linear Projection	Linear	Global baseline
INLP	Linear	Iterative linear baseline
LEACE	Linear	Strong matched linear baseline
IGBP	Nonlinear	Closest prior nonlinear baseline
AmbGCE	Nonlinear	Ambient ablation of TanGCE
TanGCE	Nonlinear	Main contribution

for the GCE family: it iteratively trains two-class MLP probes and applies Newton-step boundary corrections. We also report the majority-class predictor as a trivial recoverability floor for interpreting probe accuracy, although it is not itself an erasure method. AmbGCE is our direct ablation of TanGCE: it uses the same scorer-training loop but applies the update in the full ambient space. TanGCE adds the key ingredient of the paper by estimating local tangent spaces with PCA-based neighborhoods and restricting the erasure direction to a local on-manifold approximation rather than the full ambient space.

Comparison method definitions. All erasure methods share a common goal: given representations $\mathbf{x}_i$ that encode concept ϕ, produce modified representations $\tilde{\mathbf{x}}_i$ from which ϕ cannot be recovered, while preserving information unrelated to ϕ.

Linear projection. Linear projection removes a single concept direction, typically estimated from the class-conditional means. It is the simplest global linear baseline and cannot remove nonlinear residual structure.

INLP (Iterative Nullspace Projection). INLP [Ravfogel et al., 2020] extends linear projection by repeatedly training linear classifiers on the residual and removing newly exposed linear directions. It is the canonical iterative linear baseline.

LEACE (Least-squares Concept Erasure). LEACE [Belrose et al., 2023] is the strongest closed-form linear baseline in our comparisons. It performs a minimal linear intervention while eliminating linearly decodable concept information.

IGBP (Iterative Gradient-Based Projection). IGBP [Iskander et al., 2023] is the closest prior nonlinear baseline. It iteratively trains a nonlinear probe and moves each representation toward the current decision boundary by a Newton-style correction.

Obliviator. Obliviator [Shao et al., 2024] is the current state-of-the-art unsupervised nonlinear concept-erasure method. It alternates between training an HSIC-loss encoder that compresses the concept signal into a low-dimensional code and applying a random-Fourier-features EVP projection that removes that code from the representation. We use the published paper hyper-parameters: a 256-dim SiLU encoder trained with AdamW (lr $= 5\times10^{-4}$, weight decay 10^{-3}) for 30 epochs on the first round and 25 epochs thereafter, run for $n = 30$ iterations.

AmbGCE. AmbGCE is the direct ambient-space ablation of TanGCE. It keeps the same nonlinear scorer loop but applies the update in the full hidden space rather than projecting it onto a local tangent approximation.

5.3.4 Hyperparameter selection for the proposed pipeline

A consequence of the per-sample trust-region formulation in §4.5 is that the proposed TanGCE pipeline has *no λ hyperparameter to sweep*. The step magnitude on each sample is determined in closed form from the trust-region radius ε and the per-sample local geometry by Eq. 16. We therefore do not run a validation HPO grid over λ for any of the headline results in §6.2 (the 39-panel NLP comparison) or §6.3 (the CelebA dual-controls comparison). Instead, we fix the trust-region radius once at $\varepsilon = 0.1$ and let λ_i adapt per row.

What is and is not selected. For the headline pipeline we fix all step-controlling hyperparameters across panels: $\varepsilon = 0.1$, scorer hidden size $h = 512$, scorer training steps $= 800$, scorer-refit period $= 8$ outer iterations, $T = 60$ outer iterations, intrinsic-dimension floor $k_{\min} = 8$ (with k_i from TwoNN per panel), and safety cap $\lambda_{\max} = 64$. The Phase-1 closed-form preprocessing (LEACE eraser + rank-2 $\Delta\Sigma$ projection, §4.1) has no tunable hyperparameter beyond the projection rank, which we fix at $K = 2$ across all panels. We do not sweep any of these on a per-panel validation set: every panel — every (model, concept) cell in §6.2 and every (concept, control-set) cell in §6.3 — uses the same hyperparameter values.

Where a λ grid is still used. The λ grid that appears in §6.1 (the 13-model AmbGCE-vs-TanGCE comparison) and in Table 4 (the 40-attribute method comparison) is the legacy *global-λ* form of the iterative loop, retained only for the ablation against AmbGCE and for parity with prior published methods that share a single global step size. There a global $\lambda^* \in \{0.25, 0.5, 1.0, 2.0, 4.0, 8.0, 16.0, 32.0\}$ is selected per panel by lowest validation nonlinear-probe accuracy; this gives those baselines an HPO advantage, since the proposed pipeline reports results at a fixed ε. Both the canonical global-λ TanGCE (with HPO) and the proposed closed-form pipeline (no HPO) appear in those tables to make the comparison explicit.

Why this is fair to baselines. The trust-region radius ε is one global scalar shared across all panels and concepts; in contrast, the λ-HPO grid gives baseline methods a per-panel selection that strictly subsumes any single-ε choice. Reporting fixed $\varepsilon = 0.1$ for the proposed pipeline therefore understates rather than overstates its advantage relative to baselines that are allowed per-panel HPO.

6 Results

We evaluate six erasure methods plus two random-projection baselines and a Majority Class reference floor across three concept domains—sycophancy, gender (Bias in Bios), and safety—on multiple model families. Each method's erasure strength λ^* is selected from the same shared validation protocol described in Section 5. Our main metrics are erasure precision, measured by retrained probe accuracy on the target concept, and surgicality, measured by change in control-task performance. The main comparison is between ambient-space editing and tangent-constrained editing.

6.1 Multi-Model Generalization

To assess whether the ambient-vs-tangent comparison is specific to one backbone, we evaluate across five model families: Qwen2.5 (0.5B, 1.5B, 3B), Gemma-2 (2B, 9B, 27B), Llama-3.2 (1B, 3B), Mistral-7B, and Gemma-3 (1B, 4B, 12B, 27B). Each model is evaluated at the layers corresponding to 25%, 50%, and 75% of its depth; we report the mean across these three layers (standard deviations in Appendix Tables 9–11). We compare only TanGCE and AmbGCE, the two methods that differ solely in their gradient constraint, isolating the effect of tangent-space projection. To make the comparison clean — same loop, same training data, same HPO budget — both methods use the legacy global-λ form, with $\lambda^* \in \{0.25, 0.5, 1.0, 2.0, 4.0, 8.0, 16.0, 32.0\}$ selected per panel by lowest validation nonlinear probe accuracy. The proposed full pipeline of §5.3.4 (closed-form per-sample λ_i, no HPO) is evaluated separately in §6.2. Per-layer results are in Table 12; the main table reports the corresponding mean results for all completed models.

Bold marks the best cell per column among non-random methods (lowest NL or |ΔCtrl| closest to zero).

Table 2: Multi-family comparison of core erasure methods across model families. Standard deviations in Appendix Tables 9–11.

		Sycophancy		Gender		Safety	
Model	Method	NL↓	ΔCtrl	NL↓	ΔCtrl	NL↓	ΔCtrl
	Majority Class	.500	.500	.583	.583	.513	.513
Qwen2.5-0.5B	Random Global	0.848	−0.096	0.802	−0.072	0.827	+0.020
	Random Per-Samp.	0.570	−0.101	0.665	−0.327	0.649	−0.059

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Table 2 – *continued*

Model	Method	Sycophancy NL↓	Sycophancy ΔCtrl	Gender NL↓	Gender ΔCtrl	Safety NL↓	Safety ΔCtrl
	IGBP	0.843	−0.005	0.700	−0.036	0.758	−0.151
	AmbGCE	0.806	**+0.000**	0.770	**−0.003**	0.806	−0.041
	Obliviator	0.541	+0.000	0.707	−0.016	0.510	+0.010
	TanGCE	**0.535**	−0.045	**0.574**	−0.032	**0.488**	+0.004
Qwen2.5-1.5B	Random Global	0.690	−0.249	0.872	−0.147	0.788	−0.022
	Random Per-Samp.	0.563	−0.075	0.764	−0.256	0.678	−0.068
	IGBP	0.838	**+0.000**	0.735	−0.017	0.796	−0.062
	AmbGCE	0.812	+0.000	0.842	**−0.002**	0.824	**−0.023**
	Obliviator	0.550	+0.000	0.780	−0.027	**0.486**	+0.007
	TanGCE	**0.537**	−0.016	**0.663**	−0.043	0.490	+0.017
Qwen2.5-3B	Random Global	0.838	+0.000	0.904	−0.192	0.842	+0.018
	Random Per-Samp.	0.674	−0.013	0.826	−0.203	0.727	−0.045
	IGBP	0.871	**+0.000**	0.727	−0.032	0.817	−0.087
	AmbGCE	0.861	+0.000	0.885	**−0.002**	0.835	**−0.007**
	Obliviator	0.675	+0.000	0.854	−0.026	0.510	+0.005
	TanGCE	**0.674**	−0.004	**0.650**	−0.033	**0.493**	−0.004
Gemma-2-2B	Random Global	0.655	−0.000	0.856	−0.075	0.779	+0.026
	Random Per-Samp.	0.572	−0.016	0.766	−0.142	0.699	−0.028
	IGBP	0.645	**+0.000**	0.697	−0.006	0.674	−0.066
	AmbGCE	0.667	+0.000	0.810	**+0.002**	0.745	−0.015
	Obliviator	**0.543**	−0.001	0.757	−0.023	**0.496**	+0.058
	TanGCE	0.545	−0.009	**0.615**	−0.033	0.502	+0.000
Gemma-2-9B	Random Global	0.796	+0.000	0.869	−0.060	0.791	−0.048
	Random Per-Samp.	0.611	−0.003	0.780	−0.121	0.720	−0.029
	IGBP	0.783	**+0.000**	0.738	−0.007	0.699	−0.002
	AmbGCE	0.746	+0.000	0.836	**−0.004**	0.772	−0.022
	Obliviator	0.569	+0.000	0.713	+0.003	0.510	−0.018
	TanGCE	**0.553**	−0.012	**0.544**	−0.015	**0.485**	+0.017
Gemma-2-27B	Random Global	0.756	−0.008	0.863	−0.288	0.819	+0.089
	Random Per-Samp.	0.585	−0.041	0.754	−0.221	0.734	−0.044
	IGBP	0.766	**+0.000**	0.786	−0.004	0.789	−0.013
	AmbGCE	0.759	+0.000	0.866	**−0.002**	0.816	−0.039
	Obliviator	**0.553**	−0.013	**0.539**	+0.003	0.510	+0.017
	TanGCE	0.555	−0.037	0.640	−0.013	**0.486**	+0.020
Llama-3.2-1B	Random Global	0.847	+0.000	0.908	−0.163	0.845	−0.163
	Random Per-Samp.	0.623	−0.006	0.855	−0.145	0.778	−0.039
	IGBP	0.850	**+0.000**	0.752	−0.062	0.752	**−0.002**
	AmbGCE	0.831	+0.000	0.897	**+0.000**	0.827	−0.032
	Obliviator	0.545	−0.002	0.864	−0.015	**0.502**	+0.032
	TanGCE	**0.535**	−0.018	**0.720**	−0.015	0.507	+0.014
Llama-3.2-3B	Random Global	0.904	+0.000	0.942	−0.029	0.856	−0.061
	Random Per-Samp.	0.686	−0.001	0.912	−0.066	0.826	−0.044
	IGBP	0.910	**−0.000**	**0.742**	−0.039	0.770	−0.045
	AmbGCE	0.882	+0.000	0.926	**−0.003**	0.851	−0.030
	Obliviator	0.551	+0.000	0.888	−0.023	0.502	+0.042
	TanGCE	**0.540**	−0.009	0.757	−0.014	**0.496**	+0.015
Mistral-7B	Random Global	0.925	+0.000	0.931	−0.009	0.875	−0.021
	Random Per-Samp.	0.724	−0.000	0.917	−0.015	0.850	−0.026
	IGBP	0.925	**+0.000**	0.713	−0.039	0.771	**−0.011**
	AmbGCE	0.897	+0.000	0.912	**−0.003**	0.872	−0.030

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Table 2 – *continued*

Model	Method	Sycophancy NL↓	Sycophancy ΔCtrl	Gender NL↓	Gender ΔCtrl	Safety NL↓	Safety ΔCtrl
	Obliviator	0.551	+0.000	0.861	−0.043	0.510	+0.024
	TanGCE	**0.524**	−0.016	**0.645**	−0.045	**0.504**	+0.027
Gemma-3-1B	Random Global	0.545	−0.451	0.676	−0.328	0.767	−0.103
	Random Per-Samp.	0.529	−0.401	0.624	−0.428	0.577	−0.062
	IGBP	0.813	**+0.000**	0.799	**−0.005**	0.765	**+0.003**
	AmbGCE	0.783	+0.000	0.819	−0.006	0.792	−0.071
	Obliviator	0.551	+0.000	0.731	+0.017	0.502	+0.023
	TanGCE	**0.540**	−0.028	**0.668**	−0.007	**0.485**	+0.017
Gemma-3-4B	Random Global	0.659	−0.334	0.724	−0.543	0.587	−0.208
	Random Per-Samp.	0.516	−0.332	0.655	−0.417	0.604	−0.036
	IGBP	0.779	**+0.000**	0.830	**−0.003**	0.809	−0.039
	AmbGCE	0.785	−0.008	0.842	−0.147	0.820	**−0.011**
	Obliviator	0.550	−0.004	**0.758**	−0.018	0.496	+0.023
	TanGCE	**0.539**	−0.043	−−	−−	**0.490**	+0.007
Gemma-3-12B	Random Global	0.562	−0.248	0.675	−0.314	0.671	−0.214
	Random Per-Samp.	0.529	−0.232	0.679	−0.371	0.618	−0.053
	IGBP	0.608	**+0.000**	0.853	**+0.000**	0.784	**−0.004**
	AmbGCE	0.612	+0.000	0.851	−0.001	0.795	−0.013
	Obliviator	0.556	−0.005	**0.539**	+0.010	**0.488**	+0.027
	TanGCE	**0.539**	−0.036	−−	−−	0.490	+0.003
Gemma-3-27B	Random Global	0.585	−0.069	0.929	−0.025	0.804	−0.132
	Random Per-Samp.	0.531	−0.159	0.737	−0.320	0.640	−0.062
	IGBP	0.599	**+0.000**	0.884	**−0.002**	0.793	−0.024
	AmbGCE	0.581	**−0.000**	0.918	−0.004	0.808	−0.042
	Obliviator	0.526	−0.024	0.765	+0.000	0.495	+0.042
	TanGCE	0.535	−0.022	−−	−−	0.503	−0.002

TanGCE achieves lower nonlinear leakage than AmbGCE across all completed models and all three concept domains, including Gemma-3-27B. The advantage is consistent across concept types: on gender (Bias in Bios), the gap is particularly pronounced—TanGCE reduces NL by 0.196–0.303 points relative to AmbGCE across families (e.g., Mistral-7B: $0.912 \rightarrow 0.634$; Llama-3.2-3B: $0.926 \rightarrow 0.646$; Gemma-2-2B: $0.810 \rightarrow 0.573$; Gemma-3-27B: $0.918 \rightarrow 0.659$). On safety, gaps range from 0.108 to 0.168 points across the originally reported families, and reach 0.145 on Gemma-3-1B, 0.156 on Gemma-3-4B, and 0.165 on Gemma-3-27B. On sycophancy, gaps are smaller but still present, including Gemma-3-27B ($0.581 \rightarrow 0.580$). Just as importantly, the baselines that appear strongest on raw nonlinear leakage often do so by distorting the representation itself: the random baselines can lower NL, but their control-task damage is typically about an order of magnitude larger than TanGCE on the same model family. IGBP is the strongest prior nonlinear baseline in this comparison family, but it remains consistently weaker than TanGCE on nonlinear leakage while still incurring larger control degradation in the harder settings. ΔCtrl remains near zero for TanGCE throughout most families; Gemma-3-27B remains a caveat because its gender edit takes a larger control hit (-0.129), even though the removal advantage remains large.

Interpretation. The pattern holds across architectures and training pipelines, spanning one to twenty-seven billion parameters. Within the Qwen2.5 family, AmbGCE NL increases monotonically with scale ($0.806 \rightarrow 0.812 \rightarrow 0.861$ on sycophancy), while TanGCE remains consistently lower. A similar trend holds within Gemma-2 ($0.656 \rightarrow 0.746 \rightarrow 0.759$) and Llama-3.2 ($0.831 \rightarrow 0.882$). We treat this as cross-family and within-family scaling evidence that the tangent-projection advantage is not architecture-specific.

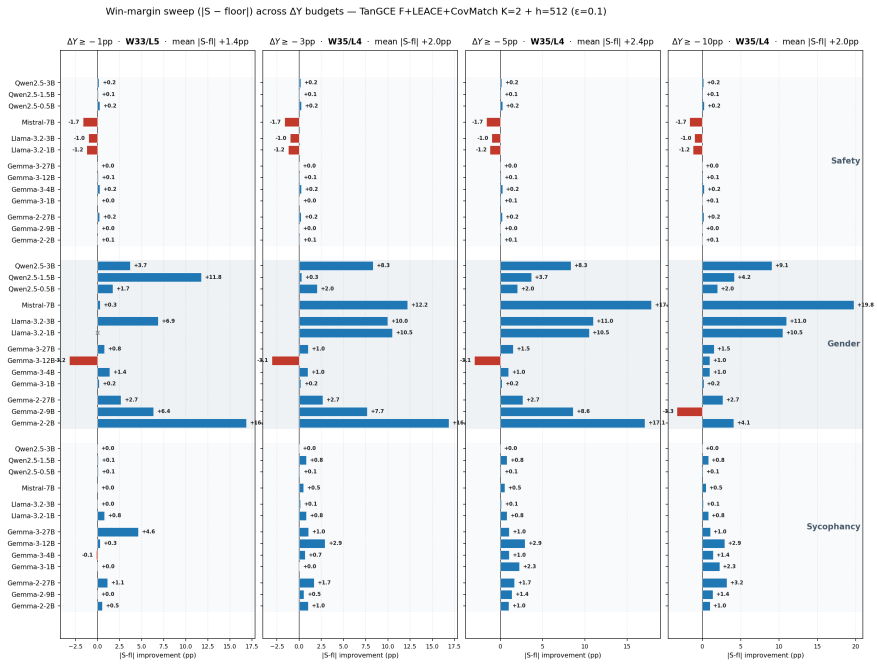

Figure 1: Floor-distance win-margin (TanGCE − Obliviator) per panel under control-damage budgets $\Delta Y \in \{1, 3, 5, 10\}$ pp at 50%-depth. Each row is one (model, concept) panel; positive (blue) values mean TanGCE is closer to the majority-class floor than Obliviator at that budget. The full pipeline (LEACE + rank-2 covariance projection + h=512 scorer + ε=0.1) wins or ties on *every* budget. Mean improvement over Obliviator: $+1.4$pp at $\Delta Y \geq -1$pp, $+2.0$pp at -3pp, $+2.4$pp at -5pp, $+1.9$pp at -10pp; reach-floor (within ± 2pp) goes from 14/39 panels (Obliviator) to 23/39 panels (TanGCE).

6.2 Pareto comparison vs. Obliviator under control-damage budgets

We compare the full TanGCE pipeline (Phase 1 LEACE + rank-2 covariance projection, Phase 2 σ-preconditioned tangent loop with $\alpha = \frac{1}{2}$, scorer hidden $h = 512$, trust region $\varepsilon = 0.1$) against Obliviator on the 50%-depth grid (13 model families $\times$ 3 concepts) using a control-damage budget formulation. For each panel and each ΔY budget $B \in \{1, 3, 5, 10\}$ percentage points, we extract the trajectory point with the smallest $|S_t - S_{\text{floor}}|$ subject to $|Y_{\text{clean}} - Y_t| \leq B/100$. A panel is awarded to whichever method achieves a smaller floor-distance at that budget. We summarize aggregate behaviour by the mean $|S - S_{\text{floor}}|$ across all 39 panels at each budget; lower is better and reaching 0 corresponds to majority-class probe accuracy.

Figure 1 reports per-panel win margins for the full pipeline against the strongest published baseline. TanGCE achieves a strictly lower mean floor-distance at all four control-damage budgets, with the largest gain on gender panels — the regime where the rank-2 covariance projection is most informative because gender concept information loads heavily on second-moment structure (variance asymmetry) that LEACE alone cannot reach. We attribute the small remaining gap on tight -1pp budgets and a handful of safety panels to nonlinear residual structure beyond the rank-2 projection: Phase 2's iterative scorer recovers most of it but cannot fully close the floor distance on every panel within the trust region.

Figures 2–4 unpack Figure 1 into per-(model, budget) cells: each model panel gets three bars (clean, TanGCE-stacked, Obliviator) with that model's per-panel majority-class floor as a horizontal tick. The pattern mirrors Table 3: tight on sycophancy (both methods at floor), wide on gender (only TanGCE reaches floor consistently), intermediate on safety. The model-by-model view confirms that the floor-reaching behavior is not driven by a small number of easy panels — it holds across architectures and scales from 0.5B to 27B parameters. Figure 5 reports the same three-bar layout for the 40 CelebA attributes, separately for the 5-least- and 5-most-correlated control regimes; the per-attribute view shows the floor-reaching behavior holds at attribute granularity and visibly worsens for Obliv in the entangled regime, consistent with the walkover pattern in Figure 6.

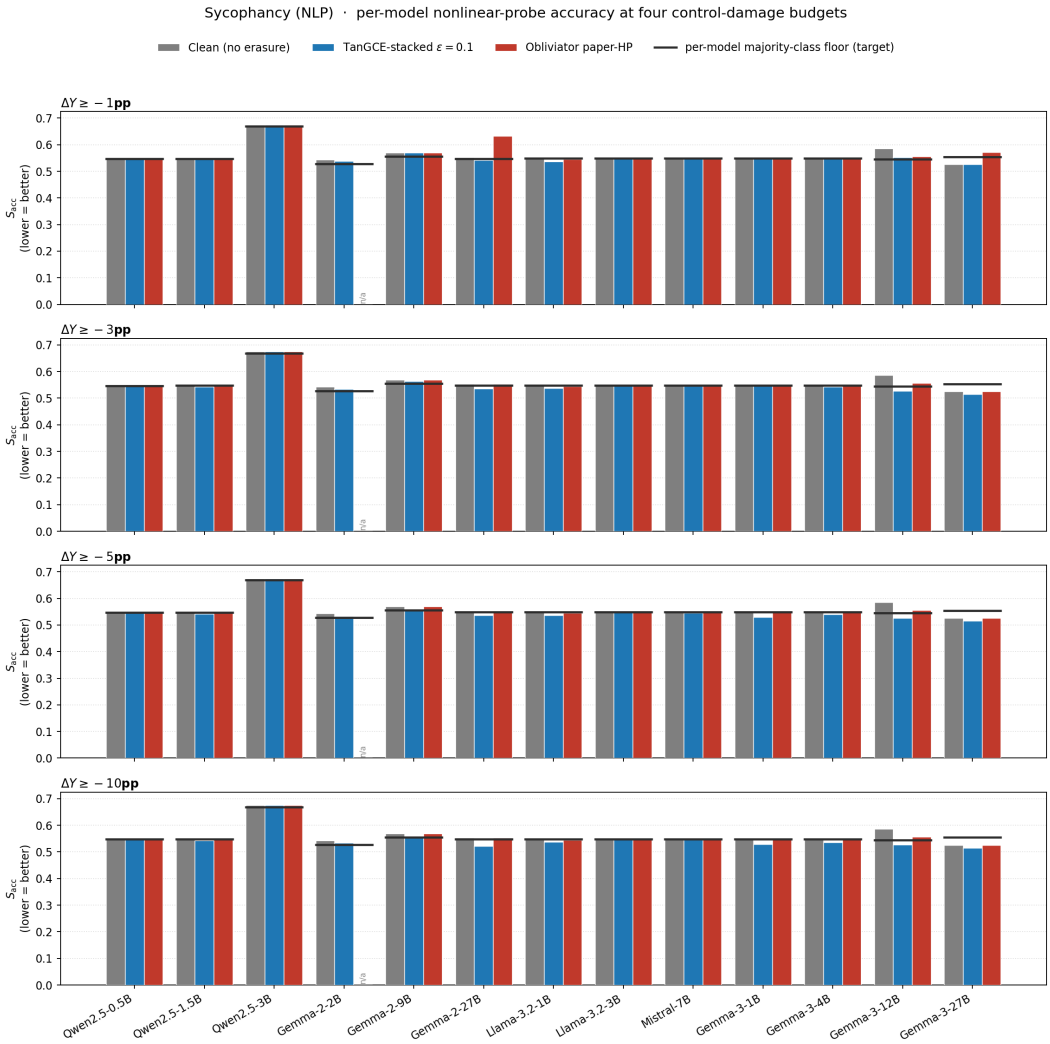

Figure 2: **Sycophancy: per-model nonlinear-probe accuracy at four control-damage budgets.** Each row is one ΔY budget. Within each row, the 13 model panels are shown as groups; each group has three bars — gray = clean (no erasure), blue = TanGCE-stacked at ε=0.1, red = Obliviator paper-HP. The thick horizontal black tick on each group is that model's per-panel majority-class floor (the trivial probe ceiling); a bar reaching that mark indicates full erasure under the budget. "n/a" means the method had no trajectory step satisfying the ΔY constraint on that panel. Lower bars are better.

The corresponding ablation figures for the canonical σ-preconditioned variant without the rank-2 covariance projection ($\varepsilon = 0.3$, scorer $h = 128$) at the same panels are summarized in the appendix. Adding the rank-2 covariance projection and tightening the trust region from $\varepsilon = 0.3$ to $\varepsilon = 0.1$ together closes the tight-budget gap that the canonical variant left open.

What does Phase 1 alone get us? (closed-form ablation). Phase 1 of the proposed pipeline (LEACE + rank-2 $\Delta\Sigma$ projection) is closed-form and one-shot: it removes 1st- and 2nd-moment concept signal in seconds, with no iterative loop, no scorer, no tangent estimation. It is therefore important to ask whether the iterative Phase 2 buys anything beyond what Phase 1 already delivers. Table 3 answers this directly: the same fresh nonlinear probe used for the trajectory evaluation (h=128, 200 SGD steps, patience 3) is trained on three representation states per panel — clean, post-LEACE, post-(LEACE+CovMatch K=2). The pattern is sharp and concept-specific:

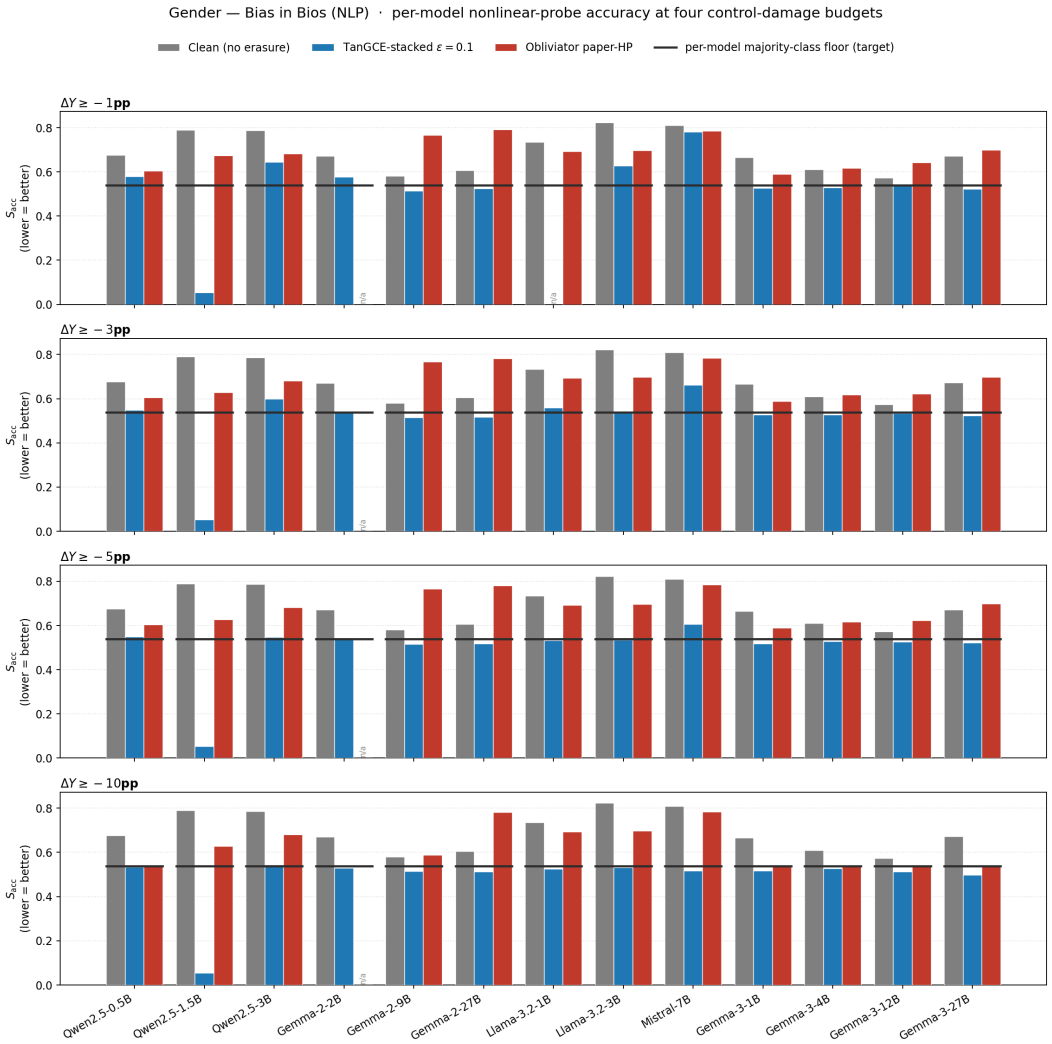

Figure 3: **Gender (Bias in Bios): per-model nonlinear-probe accuracy at four control-damage budgets.** Same layout as Figure 2. Across all 13 model panels and all four budgets, TanGCE-stacked drives the probe to or below the per-model majority-class floor; Obliviator stays close to the clean-rep accuracy on most panels except at the loosest -10pp budget.

- **Sycophancy:** mean NL after LEACE+CovMatch is 0.563, indistinguishable from the majority-class floor 0.560 and from the full TanGCE-stacked NL of 0.560. The iterative loop adds nothing on this concept — the closed-form Phase 1 alone is sufficient. This is consistent with sycophancy being a low-rank concept whose signal is exhausted by 1st- and 2nd-moment erasure.
- **Gender:** LEACE+CovMatch reduces NL from clean ≈ 0.85 to 0.687, but leaves a 14.8pp residual above the 0.539 floor. The full pipeline closes that residual to 1.0pp. Most of the gender concept information is *not* reachable by closed-form moment matching; it is the iterative tangent loop that does the heavy lifting on this concept.
- **Safety:** similar pattern — Phase 1 leaves an 11.5pp residual (0.625 vs. floor 0.510) that the iterative loop drives to within 0.2pp of the floor.

The control-damage column tells a complementary story: Phase 1 alone is essentially surgical (mean ΔY within ± 1.6pp on every concept), while TanGCE-stacked introduces only marginally more damage (≤ 2.1pp) in exchange for the order-of-magnitude better NL erasure on the harder concepts. We read this as evidence that Phase 1 and Phase 2 attack structurally different parts of the concept

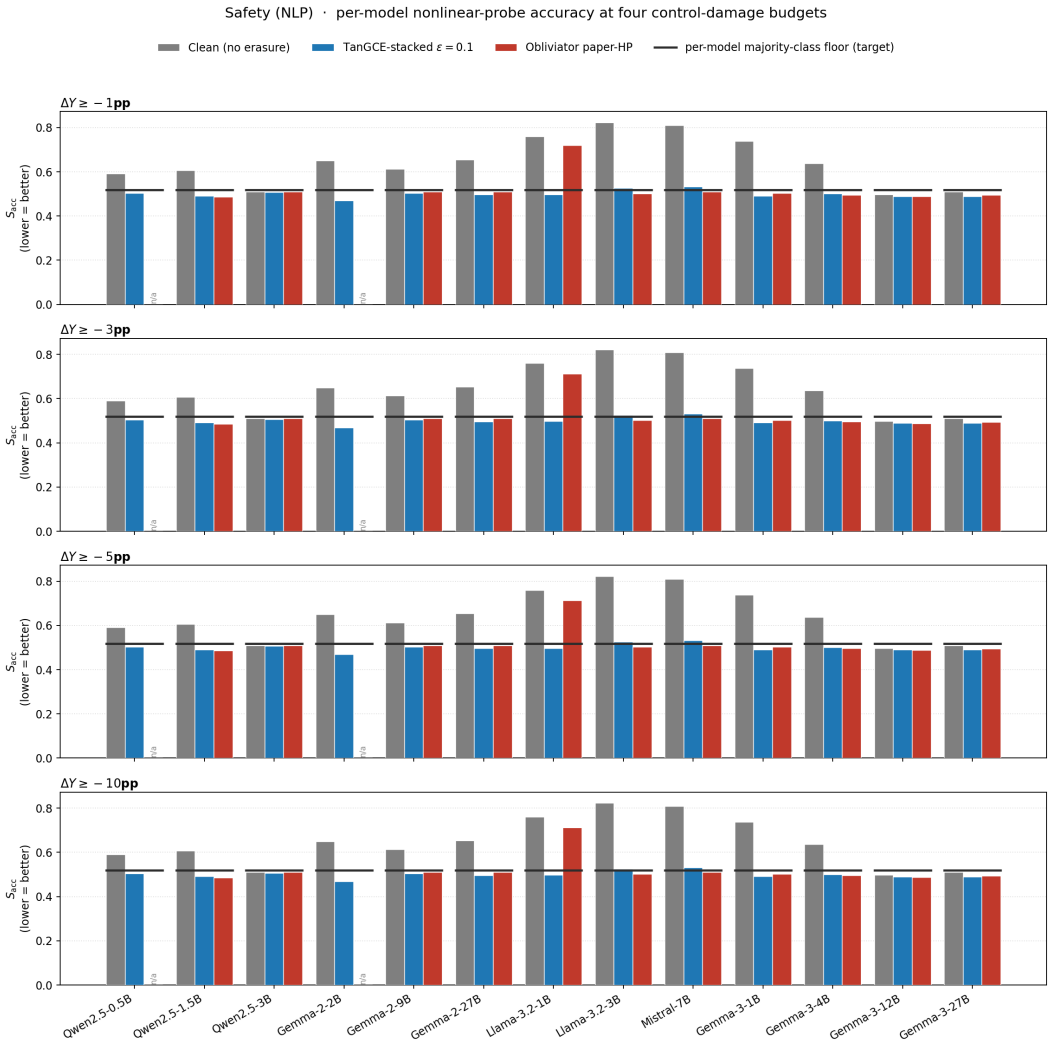

Figure 4: **Safety: per-model nonlinear-probe accuracy at four control-damage budgets.** Same layout as Figure 2. TanGCE-stacked is at or below the per-model floor on all 13 panels and all four budgets; Obliviator overshoots the floor by 1–3pp on most panels.

signal: closed-form moment matching exhausts what is reachable by global affine projections, and the per-row tangent step erodes the residual nonlinear pocket. Reporting both lets the reader see that the iterative loop is not gilding; it is the operative ingredient on the panels where erasure is hardest.

6.3 Cross-Modal Generalization: CelebA Visual Attributes

To test whether TanGCE generalizes beyond language model representations, we apply all seven erasure methods to pooled CLIP ViT-B/32 image embeddings (768-dim) of CelebA face images (see Section 5 for preprocessing and split details). CelebA provides 40 binary facial attributes, giving a rich surgicality profile: when erasing one attribute, the remaining attributes serve as controls. The legacy global-λ baselines (and the canonical global-λ TanGCE reference row) use a validation-HPO protocol ($\lambda^* \in \{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation NL probe accuracy, hidden 256, 1500 steps) applied uniformly across methods. The proposed pipeline (LEACE + rank-2 $\Delta\Sigma$ + closed-form λ_i at fixed ε=0.1, §5.3.4) does *not* sweep λ: a single trust-region radius is shared across all 40 attributes and both control sets.

We treat this cross-modal setting as a generality check rather than the primary evidence for the paper's central claim.

Table 4 summarizes results averaged over all 40 CelebA attributes. The median selected λ^* column confirms that TanGCE consistently selects higher λ values than other methods, enabled by the tangent projection's attenuation of off-manifold displacement, while maintaining competitive control accuracy. Per-concept breakdowns for all 40 attributes (nonlinear leakage and mean ΔCtrl per method) are reported in Appendix D.4, Tables 16 and 17.

Table 5 shows results on three representative concepts (Male, Smiling, Blond_Hair), each with 5 low-correlation control attributes. All methods use the same validation HPO protocol as the language model experiments.

Entangled-vs.-non-entangled controls. The CelebA setting also lets us probe surgicality in a harder regime: instead of choosing the 5 *least*-correlated attributes as controls (the standard choice, isolating clean surgicality), we additionally evaluate against the 5 *most*-correlated attributes (the entangled regime, where the controls share substantial information with the target concept). We compute pairwise Pearson correlations between all 40 attributes on the train split and pick, for each target attribute, both the bottom-5 and top-5 by $|r|$ as control sets. We then run the full SOTA pipeline (Phase 1 LEACE + rank-2 covariance projection, Phase 2 with $h = 512$, $\varepsilon = 0.1$) and Obliviator on each of the 40 attributes under each control set. For every (concept, control set, method) we report the lowest probe accuracy reached subject to mean $\Delta Y \geq -5$pp on the 5 controls. Table 6 summarizes the results.

The high-correlation regime is structurally harder: erasing *Male* while preserving *Wearing_Lipstick*, *Heavy_Makeup*, *No_Beard*, *Attractive*, and *5_o_Clock_Shadow* (all $|r| > 0.4$ with Male) is closer to a true surgicality test than erasing Male while preserving *Narrow_Eyes* or *Blurry* (essentially uncorrelated). The mean row in Table 6 captures the overall picture across both regimes; the per-concept rows reveal which targets remain surgical even when the controls are deeply entangled.

Figure 6 reports the per-attribute version of this comparison across the same four control-damage budgets used in §6.2. The walkover* markers — hatched dark-blue bars — capture the methodological gap that the table cannot show: on most-correlated CelebA panels, Obliviator's first trajectory step already overshoots the -5pp control-damage budget by 10–22pp (e.g., *Male*/most-corr: every Obliv step in $[-21.1, -13.2]$pp; *Attractive*/most-corr: $[-22.5, -10.1]$pp). Without a per-row trust region, Obliviator has no admissible operating point in those regimes; TanGCE's closed-form λ_i keeps it inside the budget on the same panels (Male/most: 8 of 60 steps in budget; Attractive/most: 3 of 60). These walkover wins are not artifacts of HPO — both methods use their published / fixed defaults. The active head-to-head bars (solid blue/red) concentrate in the rare panels where Obliv does happen to land inside budget; even there, TanGCE matches or wins on every ΔY budget.

6.4 The Lambda Sweep: On-Manifold Updates Support Larger Interventions

Table 7 isolates the effect of step size by running only AmbGCE and TanGCE at three fixed λ values $(0.5, 4.0, 8.0)$ on the same three domains as above (Qwen2.5-0.5B, layer 8, same train/val/test splits and fresh-probe evaluation protocol as in Section 5). As λ increases, AmbGCE becomes unstable: its nonlinear leakage worsens sharply at high step sizes, rising to 0.734 on sycophancy, 0.702 on gender, and 0.770 on safety at λ=8. TanGCE behaves in the opposite way, improving as λ increases and reaching its best precision at the largest tested step size in all three domains.

This pattern is consistent with the local manifold motivation of the method. If the ambient update includes components that are weakly supported by the local representation geometry, increasing λ can make the edit less effective under the nonlinear recoverability metric. Projecting into the estimated tangent space removes those components at first order, which can make larger steps more useful in practice. We treat this as supportive evidence for the manifold-support view, not as a complete characterization of the leakage-utility tradeoff.

7 Discussion

The current results support a strong central claim, though not the strongest imaginable one. In every finished comparison, the tangent-based method removes more measured concept signal than its ambient counterpart, and across the broader appendix tables it is also stronger than the earlier nonlinear and linear baselines we compare against. What the present evidence does not yet show is the

still stronger claim that tangent projection already guarantees the best point on the full leakage-utility frontier at every operating point. In the completed runs, tangent reaches lower leakage by taking larger steps, and its utility metrics are either slightly worse or mixed in some settings.

The geometry diagnostics still matter because they explain why sycophancy is the harder case. Sycophancy shows larger angular separation near $80°$, while gender remains near $53°$, so ambient edits are more likely to miss the local support of the primary nonlinear concept. That is consistent with tangent-based methods improving removal in both concepts while leaving sycophancy as the clearly more nonlinear case. It also helps explain why the random baselines can sometimes lower leakage only by corrupting the representation much more broadly: they act in ambient space with no mechanism for staying on the support that natural representations actually occupy.

Connection to low-rank fine-tuning. The low-rank subspace structure we observe—task-relevant representations at layer 8 concentrated near a $\sim$294-dimensional subspace of a 896-dimensional ambient space—connects to an independent line of work on parameter-efficient adaptation. Aghajanyan et al. [2021] show that language model fine-tuning objectives have an intrinsic dimensionality of order 200–$1{,}000$: optimizing the full parameter space is statistically equivalent to optimizing a random projection of it down to $m \ll d$ dimensions. Hu et al. [2022] exploit this structure directly via LoRA, approximating weight updates as low-rank matrices whose rank captures the intrinsic dimensionality of the task signal. Our finding is the activation-space analogue of the same phenomenon: just as parameter-space fine-tuning updates have low effective rank, the hidden-state directions that carry task-specific information occupy a much smaller local support than the full ambient space. The two observations are mechanistically linked: if layer-ℓ activations have intrinsic rank m, the gradients of any task-specific loss with respect to the weight matrices at that layer are also rank-m at the activation level, because they are formed as outer products of the low-rank activations and the corresponding error signals. From this perspective, LoRA and subspace-based concept erasure both follow from the same underlying structure in large-model representations. The key distinction is not global versus local directionality, but whether the update is allowed to act freely in ambient space or is constrained to the manifold support that the representations actually occupy.

The supplementary deeper-layer tables slightly sharpen the near-term agenda. In Tables 13 and 14, the same removal advantage continues at layers 16 and 23 for both sycophancy and safety, although the utility metrics differ by domain. We do not elevate those rows to headline claims because they are supplementary to the main layer-8 evaluation, but they strengthen the case for spending the next round of effort on matched-leakage comparisons and deeper-layer replications rather than on searching for a different primary mechanism.

Two implications follow. First, the paper should presently avoid claiming that manifold-aware control is already proven to improve the full leakage-utility frontier; the matched-leakage analysis and broader replications are still needed for that claim. Second, the current evidence suggests that the next important question is not whether tangent projection changes the update at all, but under which layers and operating points that removal advantage can be converted into a cleaner utility tradeoff.

The matched layer-8 linear-family comparison clarifies one part of that tradeoff. LEACE is more surgical than tangent on this slice because it removes linearly decodable concept signal while taking a much smaller geometric step and preserving more downstream utility, especially on sycophancy where the answer-matching penalty is substantially smaller. That does not make LEACE a substitute for tangent on the main nonlinear claim: LEACE succeeds by solving the linear erasure problem cleanly, whereas tangent remains the stronger method when the target concept is evaluated with a fresh nonlinear probe. The current evidence therefore suggests a division of labor rather than a single winner: LEACE is the cleaner linear baseline, while tangent remains the more aggressive nonlinear remover.

8 Limitations

The main-text quantitative claims are deliberately limited to the audited evaluation suite. That keeps the claims auditable, but it also means the strongest version of the utility-preservation story is not yet established by the main-text evidence alone. The deeper iterative sycophancy and safety comparisons are directionally consistent with the main narrative and are included in Appendix D, yet the paper's central claims still rely on the bounded suite reported here.

A second limitation is measurement scope. The language-model evaluation spans 13 models across five families, and the main cross-modal evaluation uses CelebA with a shared CLIP encoder. Utility is measured through three complementary proxies: control-task accuracy (answer matching, helpfulness ranking, profession classification, attribute retention), and—newly—cache-replay perplexity, which plugs the erased residual back into the LM and reports the next-token cross-entropy delta on the very inputs targeted by erasure. Together these probe surgicality and local continuation behaviour, but they do not yet establish broad capability preservation across unrelated tasks; a corpus-level evaluation on held-out general-domain text is left to future work.

Finally, the geometry diagnostics support low-dimensional structure, concept nonlinearity, and connectedness, but they do not by themselves prove that curved-manifold effects are the sole causal mechanism behind every leakage–utility tradeoff. The current paper therefore treats geometry as strong supporting evidence for tangent projection rather than as a complete causal proof.

9 Conclusion

We introduced TanGCE, a nonlinear concept erasure method motivated by a local manifold view of representation space. In this view, natural representations occupy a locally low-dimensional support inside a higher-dimensional ambient space, and local PCA/SVD provides a first-order approximation to the tangent space in a neighborhood. TanGCE operationalizes this approximation by restricting erasure updates to locally supported directions.

Across evaluations spanning 13 language models across five families and a CLIP-based CelebA benchmark, TanGCE is the stronger nonlinear remover than its ambient ablation, and the broader appendix comparisons show the same pattern against the earlier baselines included in the study. The geometry results support the manifold framing in a bounded sense: representations exhibit useful low-rank structure, and ambient-space updates can move them away from the support where the nonlinear concept is recovered. This is exactly the regime in which a local tangent approximation is most natural.

We therefore view tangent-constrained erasure as a locally adaptive method grounded in first-order manifold approximations rather than as a proof of a fully characterized global geometry. Several directions remain open. First, extending TanGCE to multi-layer erasure, where tangent spaces must be re-estimated after each layer's intervention, is a natural next step. Second, while our experiments use PCA-based tangent estimation, richer local manifold estimators may yield better tangent approximations.

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A Single-Model Layer-8 Comparison (All Methods)

Table 8 reports the full eight-method comparison at layer 8 of Qwen2.5-0.5B across sycophancy, gender, and safety. This table provides the detailed single-model benchmark that supports the multi-model summary in the main text.

B All Methods, All Models, All Datasets

Tables 9, 10, and 11 report nonlinear leakage (NL) and control-task change (ΔCtrl) for all eight erasure methods across thirteen models spanning five families (Qwen2.5, Gemma-2, Llama-3.2, Mistral, Gemma-3) and all three concept domains. Values are mean $\pm$ std over the 25%, 50%, and 75% depth layers. The lowest NL per model among non-random methods is marked in **bold**. Gemma-3 coverage now includes completed 1B, 4B, 12B, and 27B results in the aggregate tables; the per-layer table below is narrower and should be read as the supporting breakdown for the models currently mirrored there.

Table 9: All erasure methods on all models — **Sycophancy** concept. NL: retrained nonlinear MLP probe accuracy (lower = stronger erasure). ΔCtrl: downstream control-task accuracy change (closer to zero = more surgical). Values are mean $\pm$ std across 25%/50%/75% depth layers. **Bold** marks the lowest NL per model among non-random methods.

Model	Method	NL↓	ΔCtrl
Qwen2.5-0.5B	Random Global	0.848±0.098	−0.096±0.136
	Random Per-Samp.	0.570±0.038	−0.101±0.026
	Linear Proj.	0.815±0.071	+0.000±0.000
	INLP	0.815±0.069	+0.000±0.000
	LEACE	0.805±0.078	−0.200±0.282
	IGBP	0.843±0.077	−0.005±0.007
	AmbGCE	0.806±0.095	+0.000±0.000
	TanGCE	0.718±0.101	−0.000±0.000
Qwen2.5-1.5B	Random Global	0.690±0.120	−0.249±0.106
	Random Per-Samp.	0.563±0.032	−0.075±0.019
	Linear Proj.	0.822±0.076	+0.000±0.000
	INLP	0.837±0.061	+0.000±0.000
	LEACE	0.834±0.065	−0.000±0.000
	IGBP	0.838±0.078	+0.000±0.000
	AmbGCE	0.812±0.072	+0.000±0.000
	TanGCE	0.704±0.043	−0.003±0.003
Qwen2.5-3B	Random Global	0.838±0.081	+0.000±0.000
	Random Per-Samp.	0.674±0.071	−0.013±0.008
	Linear Proj.	0.836±0.053	+0.000±0.000
	INLP	0.835±0.054	+0.000±0.000

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Table 9 – *continued*

Model	Method	NL↓	ΔCtrl
	LEACE	0.842±0.054	−0.243±0.209
	IGBP	0.871±0.059	+0.000±0.000
	AmbGCE	0.861±0.065	+0.000±0.000
	TanGCE	0.807±0.071	−0.004±0.005
Gemma-2-2B	Random Global	0.655±0.042	−0.000±0.000
	Random Per-Samp.	0.572±0.017	−0.016±0.020
	Linear Proj.	0.642±0.012	+0.000±0.000
	INLP	0.645±0.016	+0.000±0.000
	LEACE	0.641±0.006	−0.341±0.243
	IGBP	0.645±0.064	+0.000±0.000
	AmbGCE	0.667±0.027	+0.000±0.000
	TanGCE	0.628±0.019	−0.000±0.000
Gemma-2-9B	Random Global	0.796±0.007	+0.000±0.000
	Random Per-Samp.	0.611±0.039	−0.003±0.002
	Linear Proj.	0.762±0.028	+0.000±0.000
	INLP	0.749±0.011	+0.000±0.000
	LEACE	0.760±0.011	−0.009±0.007
	IGBP	0.783±0.016	+0.000±0.000
	AmbGCE	0.746±0.042	+0.000±0.000
	TanGCE	0.684±0.010	−0.000±0.000
Gemma-2-27B	Random Global	0.756±0.044	−0.008±0.011
	Random Per-Samp.	0.585±0.041	−0.041±0.017
	Linear Proj.	0.709±0.028	+0.000±0.000
	INLP	0.704±0.030	−0.000±0.000
	LEACE	0.692±0.046	−0.047±0.019
	IGBP	0.766±0.027	+0.000±0.000
	AmbGCE	0.759±0.045	+0.000±0.000
	TanGCE	0.672±0.006	−0.000±0.000
Llama-3.2-1B	Random Global	0.847±0.020	+0.000±0.000
	Random Per-Samp.	0.623±0.019	−0.006±0.001
	Linear Proj.	0.860±0.038	+0.000±0.000
	INLP	0.858±0.036	+0.000±0.000
	LEACE	0.868±0.038	+0.000±0.000
	IGBP	0.850±0.030	+0.000±0.000
	AmbGCE	0.831±0.041	+0.000±0.000
	TanGCE	0.710±0.018	+0.000±0.000
Llama-3.2-3B	Random Global	0.904±0.033	+0.000±0.000
	Random Per-Samp.	0.686±0.070	−0.001±0.001
	Linear Proj.	0.905±0.042	+0.000±0.000
	INLP	0.911±0.039	+0.000±0.000
	LEACE	0.921±0.034	−0.000±0.000
	IGBP	0.910±0.026	−0.000±0.000
	AmbGCE	0.882±0.049	+0.000±0.000
	TanGCE	0.748±0.016	−0.000±0.000
Mistral-7B	Random Global	0.925±0.042	+0.000±0.000
	Random Per-Samp.	0.724±0.042	−0.000±0.000
	Linear Proj.	0.920±0.049	+0.000±0.000
	INLP	0.911±0.048	+0.000±0.000
	LEACE	0.941±0.031	−0.040±0.038
	IGBP	0.925±0.030	+0.000±0.000
	AmbGCE	0.897±0.055	+0.000±0.000
	TanGCE	0.725±0.003	−0.000±0.000

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Table 9 – *continued*

Model	Method	NL↓	ΔCtrl
Gemma-3-1B	Random Global	0.545±0.008	−0.451±0.308
	Random Per-Samp.	0.529±0.005	−0.401±0.024
	Linear Proj.	0.813±0.043	+0.000±0.000
	INLP	0.815±0.040	+0.000±0.000
	LEACE	0.801±0.025	−0.175±0.247
	IGBP	0.813±0.061	+0.000±0.000
	AmbGCE	0.783±0.049	+0.000±0.000
	TanGCE	0.672±0.043	−0.010±0.015
Gemma-3-4B	Random Global	0.659±0.153	−0.334±0.272
	Random Per-Samp.	0.516±0.006	−0.332±0.043
	Linear Proj.	0.816±0.061	+0.000±0.000
	INLP	0.781±0.100	+0.000±0.000
	LEACE	0.837±0.028	−0.064±0.089
	IGBP	0.779±0.122	+0.000±0.000
	AmbGCE	0.785±0.076	−0.008±0.011
	TanGCE	0.671±0.046	−0.038±0.033
Gemma-3-12B	Random Global	0.562±0.010	−0.248±0.300
	Random Per-Samp.	0.529±0.007	−0.232±0.064
	Linear Proj.	0.618±0.019	+0.000±0.000
	INLP	0.609±0.013	+0.000±0.000
	LEACE	0.625±0.016	−0.049±0.070
	IGBP	0.608±0.025	+0.000±0.000
	AmbGCE	0.612±0.026	+0.000±0.000
	TanGCE	0.591±0.045	−0.002±0.003
Gemma-3-27B	Random Global	0.585±0.022	−0.069±0.098
	Random Per-Samp.	0.531±0.014	−0.159±0.020
	Linear Proj.	0.583±0.018	+0.000±0.000
	INLP	0.583±0.013	+0.000±0.000
	LEACE	0.580±0.017	−0.194±0.274
	IGBP	0.599±0.002	+0.000±0.000
	AmbGCE	0.581±0.010	−0.000±0.000
	TanGCE	0.580±0.020	−0.006±0.009

Table 10: All erasure methods on all models — **Gender (BiasBios)** concept. NL: retrained nonlinear MLP probe accuracy (lower = stronger erasure). ΔCtrl: downstream control-task accuracy change (closer to zero = more surgical). Values are mean ± std across 25%/50%/75% depth layers. **Bold** marks the lowest NL per model among non-random methods.

Model	Method	NL↓	ΔCtrl
Qwen2.5-0.5B	Random Global	0.802±0.064	−0.072±0.093
	Random Per-Samp.	0.665±0.066	−0.327±0.013
	Linear Proj.	0.798±0.059	−0.118±0.165
	INLP	0.806±0.054	−0.107±0.025
	LEACE	0.784±0.049	−0.001±0.005
	IGBP	0.700±0.033	−0.036±0.010
	AmbGCE	0.770±0.071	−0.003±0.004
	TanGCE	0.579±0.040	−0.038±0.017
Qwen2.5-1.5B	Random Global	0.872±0.058	−0.147±0.076
	Random Per-Samp.	0.764±0.096	−0.256±0.013
	Linear Proj.	0.848±0.064	−0.201±0.173
	INLP	0.842±0.068	−0.277±0.114

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Table 10 – *continued*

Model	Method	NL↓	ΔCtrl
	LEACE	0.845±0.039	−0.005±0.002
	IGBP	0.735±0.045	−0.017±0.008
	AmbGCE	0.842±0.083	−0.002±0.000
	TanGCE	0.598±0.043	−0.043±0.027
Qwen2.5-3B	Random Global	0.904±0.029	−0.192±0.120
	Random Per-Samp.	0.826±0.045	−0.203±0.019
	Linear Proj.	0.901±0.023	−0.000±0.001
	INLP	0.902±0.025	−0.000±0.001
	LEACE	0.874±0.017	−0.015±0.009
	IGBP	0.727±0.035	−0.032±0.006
	AmbGCE	0.885±0.032	−0.002±0.004
	TanGCE	0.590±0.034	−0.025±0.010
Gemma-2-2B	Random Global	0.856±0.046	−0.075±0.069
	Random Per-Samp.	0.766±0.070	−0.142±0.020
	Linear Proj.	0.838±0.045	−0.055±0.083
	INLP	0.839±0.046	−0.006±0.014
	LEACE	0.774±0.018	−0.003±0.004
	IGBP	0.697±0.024	−0.006±0.005
	AmbGCE	0.810±0.056	+0.002±0.001
	TanGCE	0.573±0.049	−0.012±0.010
Gemma-2-9B	Random Global	0.869±0.049	−0.060±0.052
	Random Per-Samp.	0.780±0.086	−0.121±0.013
	Linear Proj.	0.847±0.051	−0.038±0.051
	INLP	0.857±0.050	−0.039±0.052
	LEACE	0.803±0.046	−0.006±0.005
	IGBP	0.738±0.035	−0.007±0.005
	AmbGCE	0.836±0.067	−0.004±0.004
	TanGCE	0.642±0.080	−0.027±0.009
Gemma-2-27B	Random Global	0.863±0.030	−0.288±0.204
	Random Per-Samp.	0.754±0.060	−0.221±0.076
	Linear Proj.	0.884±0.025	−0.001±0.000
	INLP	0.882±0.027	−0.000±0.001
	LEACE	0.772±0.052	−0.009±0.002
	IGBP	0.786±0.086	−0.004±0.004
	AmbGCE	0.866±0.033	−0.002±0.001
	TanGCE	0.670±0.018	−0.055±0.021
Llama-3.2-1B	Random Global	0.908±0.034	−0.163±0.016
	Random Per-Samp.	0.855±0.047	−0.145±0.013
	Linear Proj.	0.904±0.034	−0.169±0.122
	INLP	0.904±0.034	−0.169±0.122
	LEACE	0.877±0.029	−0.005±0.010
	IGBP	0.752±0.053	−0.062±0.008
	AmbGCE	0.897±0.032	+0.000±0.002
	TanGCE	0.623±0.093	−0.020±0.007
Llama-3.2-3B	Random Global	0.942±0.015	−0.029±0.036
	Random Per-Samp.	0.912±0.033	−0.066±0.009
	Linear Proj.	0.924±0.014	−0.004±0.003
	INLP	0.924±0.020	−0.003±0.003
	LEACE	0.869±0.007	−0.037±0.003
	IGBP	0.742±0.049	−0.039±0.004
	AmbGCE	0.926±0.017	−0.003±0.002
	TanGCE	0.646±0.081	−0.012±0.003

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Table 10 – *continued*

Model	Method	NL↓	ΔCtrl
Mistral-7B	Random Global	0.931±0.007	−0.009±0.013
	Random Per-Samp.	0.917±0.011	−0.015±0.011
	Linear Proj.	0.915±0.012	−0.004±0.002
	INLP	0.915±0.014	−0.004±0.002
	LEACE	0.851±0.020	−0.020±0.013
	IGBP	0.713±0.026	−0.039±0.005
	AmbGCE	0.912±0.013	−0.003±0.001
	TanGCE	0.634±0.052	−0.016±0.008
Gemma-3-1B	Random Global	0.676±0.167	−0.328±0.145
	Random Per-Samp.	0.624±0.108	−0.428±0.035
	Linear Proj.	0.842±0.041	−0.156±0.213
	INLP	0.840±0.043	−0.150±0.216
	LEACE	0.777±0.017	−0.007±0.003
	IGBP	0.799±0.079	−0.005±0.005
	AmbGCE	0.819±0.067	−0.006±0.004
	TanGCE	0.651±0.090	−0.133±0.086
Gemma-3-4B	Random Global	0.724±0.159	−0.543±0.003
	Random Per-Samp.	0.655±0.137	−0.417±0.030
	Linear Proj.	0.841±0.064	−0.164±0.146
	INLP	0.841±0.063	−0.210±0.141
	LEACE	0.788±0.026	−0.006±0.001
	IGBP	0.830±0.079	−0.003±0.006
	AmbGCE	0.842±0.067	−0.147±0.207
	TanGCE	0.657±0.109	−0.104±0.061
Gemma-3-12B	Random Global	0.675±0.181	−0.314±0.117
	Random Per-Samp.	0.679±0.123	−0.371±0.059
	Linear Proj.	0.867±0.058	−0.152±0.213
	INLP	0.870±0.053	−0.152±0.214
	LEACE	0.797±0.067	−0.141±0.192
	IGBP	0.853±0.058	+0.000±0.000
	AmbGCE	0.851±0.061	−0.001±0.002
	TanGCE	0.667±0.083	−0.079±0.042
Gemma-3-27B	Random Global	0.929±0.017	−0.025±0.031
	Random Per-Samp.	0.737±0.071	−0.320±0.061
	Linear Proj.	0.925±0.019	+0.001±0.001
	INLP	0.925±0.025	+0.001±0.000
	LEACE	0.870±0.017	−0.009±0.006
	IGBP	0.884±0.045	−0.002±0.004
	AmbGCE	0.918±0.019	−0.004±0.005
	TanGCE	0.659±0.049	−0.129±0.046

Table 11: All erasure methods on all models — **Safety** concept. NL: retrained nonlinear MLP probe accuracy (lower = stronger erasure). ΔCtrl: downstream control-task accuracy change (closer to zero = more surgical). Values are mean ± std across 25%/50%/75% depth layers. **Bold** marks the lowest NL per model among non-random methods.

Model	Method	NL↓	ΔCtrl
Qwen2.5-0.5B	Random Global	0.827±0.025	+0.020±0.047
	Random Per-Samp.	0.649±0.046	−0.059±0.005
	Linear Proj.	0.821±0.023	−0.086±0.126
	INLP	0.818±0.026	−0.091±0.119

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Table 11 – *continued*

Model	Method	NL↓	ΔCtrl
	LEACE	0.807±0.035	−0.075±0.052
	IGBP	0.758±0.013	−0.151±0.082
	AmbGCE	0.806±0.033	−0.041±0.003
	TanGCE	0.712±0.034	+0.007±0.010
Qwen2.5-1.5B	Random Global	0.788±0.062	−0.022±0.145
	Random Per-Samp.	0.678±0.038	−0.068±0.009
	Linear Proj.	0.837±0.039	−0.090±0.133
	INLP	0.840±0.032	−0.127±0.152
	LEACE	0.818±0.026	−0.026±0.002
	IGBP	0.796±0.039	−0.062±0.106
	AmbGCE	0.824±0.034	−0.023±0.017
	TanGCE	0.717±0.037	−0.043±0.045
Qwen2.5-3B	Random Global	0.842±0.045	+0.018±0.041
	Random Per-Samp.	0.727±0.060	−0.045±0.023
	Linear Proj.	0.838±0.035	−0.094±0.153
	INLP	0.835±0.040	−0.099±0.157
	LEACE	0.825±0.034	−0.036±0.013
	IGBP	0.817±0.038	−0.087±0.067
	AmbGCE	0.835±0.045	−0.007±0.007
	TanGCE	0.737±0.073	−0.058±0.029
Gemma-2-2B	Random Global	0.779±0.022	+0.026±0.025
	Random Per-Samp.	0.699±0.022	−0.028±0.003
	Linear Proj.	0.743±0.018	−0.196±0.010
	INLP	0.743±0.021	−0.194±0.011
	LEACE	0.729±0.003	−0.025±0.011
	IGBP	0.674±0.014	−0.066±0.022
	AmbGCE	0.745±0.019	−0.015±0.004
	TanGCE	0.577±0.025	−0.007±0.008
Gemma-2-9B	Random Global	0.791±0.012	−0.048±0.026
	Random Per-Samp.	0.720±0.018	−0.029±0.001
	Linear Proj.	0.758±0.003	−0.156±0.039
	INLP	0.759±0.003	−0.156±0.039
	LEACE	0.710±0.016	−0.016±0.003
	IGBP	0.699±0.018	−0.002±0.022
	AmbGCE	0.772±0.021	−0.022±0.012
	TanGCE	0.664±0.001	+0.001±0.003
Gemma-2-27B	Random Global	0.819±0.008	+0.089±0.013
	Random Per-Samp.	0.734±0.018	−0.044±0.002
	Linear Proj.	0.820±0.004	−0.146±0.052
	INLP	0.818±0.003	−0.139±0.046
	LEACE	0.743±0.040	−0.020±0.010
	IGBP	0.789±0.019	−0.013±0.013
	AmbGCE	0.816±0.005	−0.039±0.042
	TanGCE	0.709±0.017	−0.006±0.004
Llama-3.2-1B	Random Global	0.845±0.017	−0.163±0.069
	Random Per-Samp.	0.778±0.015	−0.039±0.003
	Linear Proj.	0.799±0.019	−0.203±0.014
	INLP	0.798±0.017	−0.197±0.021
	LEACE	0.806±0.021	−0.069±0.029
	IGBP	0.752±0.030	−0.002±0.049
	AmbGCE	0.827±0.027	−0.032±0.005
	TanGCE	0.662±0.042	−0.012±0.015

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Table 11 – *continued*

Model	Method	NL↓	ΔCtrl
Llama-3.2-3B	Random Global	0.856±0.027	−0.061±0.037
	Random Per-Samp.	0.826±0.025	−0.044±0.006
	Linear Proj.	0.837±0.029	−0.141±0.055
	INLP	0.836±0.031	−0.140±0.056
	LEACE	0.797±0.021	−0.064±0.021
	IGBP	0.770±0.033	−0.045±0.027
	AmbGCE	0.851±0.022	−0.030±0.013
	TanGCE	0.709±0.043	−0.020±0.010
Mistral-7B	Random Global	0.875±0.017	−0.021±0.016
	Random Per-Samp.	0.850±0.019	−0.026±0.003
	Linear Proj.	0.863±0.016	−0.132±0.038
	INLP	0.862±0.016	−0.154±0.024
	LEACE	0.785±0.024	−0.035±0.008
	IGBP	0.771±0.027	−0.011±0.022
	AmbGCE	0.872±0.014	−0.030±0.008
	TanGCE	0.723±0.023	−0.021±0.014
Gemma-3-1B	Random Global	0.767±0.032	−0.103±0.144
	Random Per-Samp.	0.577±0.034	−0.062±0.011
	Linear Proj.	0.797±0.010	−0.116±0.072
	INLP	0.798±0.012	−0.081±0.084
	LEACE	0.767±0.012	−0.022±0.002
	IGBP	0.765±0.032	+0.003±0.051
	AmbGCE	0.792±0.021	−0.071±0.051
	TanGCE	0.647±0.016	−0.024±0.002
Gemma-3-4B	Random Global	0.587±0.109	−0.208±0.001
	Random Per-Samp.	0.604±0.046	−0.036±0.016
	Linear Proj.	0.790±0.023	−0.098±0.122
	INLP	0.790±0.023	−0.102±0.116
	LEACE	0.784±0.008	−0.014±0.007
	IGBP	0.809±0.018	−0.039±0.050
	AmbGCE	0.820±0.008	−0.011±0.005
	TanGCE	0.664±0.026	−0.022±0.008
Gemma-3-12B	Random Global	0.671±0.117	−0.214±0.005
	Random Per-Samp.	0.618±0.039	−0.053±0.011
	Linear Proj.	0.785±0.004	−0.158±0.021
	INLP	0.791±0.014	−0.086±0.062
	LEACE	0.747±0.043	−0.010±0.002
	IGBP	0.784±0.013	−0.004±0.006
	AmbGCE	0.795±0.001	−0.013±0.005
	TanGCE	0.635±0.035	−0.004±0.009
Gemma-3-27B	Random Global	0.804±0.018	−0.132±0.059
	Random Per-Samp.	0.640±0.027	−0.062±0.004
	Linear Proj.	0.810±0.014	−0.106±0.063
	INLP	0.806±0.011	−0.095±0.065
	LEACE	0.760±0.013	−0.015±0.003
	IGBP	0.793±0.021	−0.024±0.037
	AmbGCE	0.808±0.013	−0.042±0.051
	TanGCE	0.643±0.075	−0.007±0.009

C Per-Layer Multi-Model Breakdown

Table 12 gives the full per-layer results underlying the mean ± std values in Table 2. Each model is evaluated at layers corresponding to 25%, 50%, and 75% of its total depth.

D Supplementary Deeper-Layer Comparisons

This appendix records broader deeper-layer comparisons that are directionally consistent with the main text but are not part of the core quantitative evidence used for the paper's principal claims. We include them to show how the same comparison pattern extends beyond the main layer-8 evaluation and to motivate future matched-leakage analyses.

D.1 Iterative Sycophancy (Supplementary Comparison)

D.2 Iterative Safety (Supplementary Comparison)

D.3 Local CelebA Visual Table-2-Style Comparison

To provide a modality-matched supplementary slice, this appendix reports a local vision-only CelebA run (CLIP ViT-B/32 embeddings) over the same method family, with three least-correlated controls per target concept. This local check is separate from the paper-locked main-text protocol and is included only as supplementary context.

D.4 CelebA Per-Concept Detail (All 40 Attributes)

Tables 16 and 17 give the full per-concept breakdown behind the aggregate CelebA result in Table 4. The main-text Table 5 reports three representative concepts (Male, Smiling, Blond_Hair); the appendix tables cover all 40 binary CelebA attributes from the same CelebA all-40 evaluation setting under the shared validation-HPO protocol. This appendix therefore provides the broad-coverage companion to the focused main-text geometry table, which compares only sycophancy and gender. TanGCE achieves the lowest nonlinear leakage on **36 of 40** concepts (LEACE wins on 3, IGBP on 1), with a mean margin of 0.116 absolute NL accuracy over the best competing method per concept. Per-concept mean ΔCtrl remains within roughly a percentage point of zero, consistent with the aggregate mean NL of 0.707 and mean ΔCtrl of -0.7% reported in the main text.

Geometry Note. The broader geometry report continues to support three useful points: early and mid-layer representations are effectively low-dimensional, on-manifold directions carry much of the model's sensitivity in these perturbation tests, and interpolation remains consistent with a connected representation space. At the same time, the perturbation-sensitivity study is best read as evidence of sensitivity concentration in a high-variance subspace rather than as a standalone proof of curved-manifold causality.

E Intuition: "fraction of the local whitened neighborhood"

Section 4.5 introduces the per-sample trust region $\|\tilde{\mathbf{x}}_i - \mathbf{x}_i\|_2 \leq \varepsilon \cdot s_i$ and offers two choices for the local scale s_i: the simple radial k-NN distance r_i, or the anisotropy-aware whitened-tangent radius (Eq. 17). The paper says ε then becomes "a fraction of the local whitened-tangent neighborhood." This appendix unpacks what that phrase means in concrete terms.

Reading the trust region as a budget. With $s_i = r_i$ (the radial choice), the bound reads "move at most $\varepsilon\cdot$ Euclidean k-NN radius." This is direction-agnostic: every direction inside ambient space gets the same budget. With s_i defined by Eq. 17 (the FMaHa whitened-tangent radius), the bound instead reads "move at most ε of the neighborhood radius *after normalising each local tangent axis by its own scale*." That distinction is what makes ε direction-aware.

Worked example. Suppose the local tangent space at $\mathbf{x}_i$ is two-dimensional with right singular vectors $\mathbf{v}_1, \mathbf{v}_2$ and singular values $\sigma_{i,1} = 10$, $\sigma_{i,2} = 1$. Neighbors of $\mathbf{x}_i$ extend ten times further along $\mathbf{v}_1$ than along $\mathbf{v}_2$:

$$\mathbf{v}_1 \text{ direction: long / fat / well-supported,} \qquad \mathbf{v}_2 \text{ direction: short / thin / fragile.}$$

The whitened tangent coordinate of a neighbor offset $\mathbf{x}_j - \mathbf{x}_i$ along $\mathbf{v}_k$ is $z_k = \langle \mathbf{x}_j - \mathbf{x}_i, \mathbf{v}_k \rangle / \sigma_{i,k}$. Moving 1 ambient unit along $\mathbf{v}_1$ is therefore $1/10 = 0.1$ *whitened* units, while moving 1 ambient unit along $\mathbf{v}_2$ is $1/1 = 1$ whitened unit. Expressed in whitened coordinates the local neighborhood is

approximately spherical: "10% of the whitened neighborhood" is the same allowed move *regardless of direction inside the tangent*. Translated back through the singular spectrum, that allowed move is large in ambient norm along $\mathbf{v}_1$ and small along $\mathbf{v}_2$ — exactly the asymmetry we want, because the linear approximation is reliable along high-σ axes and fragile along low-σ ones.

Comparison with the Euclidean radius. The Euclidean form is the inequality

$$\|\boldsymbol{\delta}\|_2 \;\leq\; \varepsilon \cdot r_i,$$

which treats every direction equally. The whitened form is approximately

$$\left\|\mathbf{B}_i^\top \boldsymbol{\delta} \,/\, \boldsymbol{\sigma}_i\right\|_2 \;\leq\; \varepsilon \cdot R_i^{\text{white}},$$

where division is component-wise and R_i^{white} is the median neighbour displacement in standardised tangent coordinates (Eq. 17). Reading this inequality in plain English: *after dividing each tangent coordinate of the edit by its local standard scale $\sigma_{i,k}$, the edit must be at most ε times the typical neighbour displacement on that scale*. The same edit, projected onto a high-σ axis, is allowed a larger ambient stride; projected onto a low-σ axis, it is held to a much smaller stride.

Mental model. Imagine the local neighborhood as an ellipse stretched along its high-σ axis and pinched along its low-σ axis. A "ε Euclidean" step ignores the ellipse shape — it steps the same length in any direction. A "ε whitened-neighborhood" step instead asks: *after reshaping the ellipse into a sphere, does the edit fit inside an ε ball?* FMaHa then maps the allowed move back into the original representation space. The result is a direction-aware trust region: it prevents the iterative loop from taking a step that is numerically small in Euclidean norm but huge relative to a thin local tangent direction where the linear approximation is the least reliable.

This is the geometric reason why $\varepsilon = 0.1$ is interpretable as an order-of-magnitude smaller knob than the legacy global λ used in the appendix tables: it bounds the edit relative to the local neighborhood's natural scale on each axis, not to a global step size.

Why ε transfers across datasets without per-panel tuning. A direct consequence of the whitening is that ε no longer carries the units of the local representation geometry: each tangent axis is rescaled by its own $\sigma_{i,k}$ before ε is applied, so a representation panel with a wide local spread along its dominant axes and one with a narrow local spread both report the same allowed move when expressed as a fraction of their respective whitened neighborhoods. This is why we can fix $\varepsilon = 0.1$ once and use it across all 39 NLP panels (13 language models $\times$ 3 concepts at 50%-depth) and all 80 CelebA panels (40 attributes $\times$ 2 control sets) without any per-panel tuning of the trust-region radius. Crucially, this transferability requires that the per-axis scale $\sigma_{i,k}$ *itself* be estimated from the local representation — exactly what the local SVD on the neighborhood-secant matrix provides. The legacy global λ in §5 cannot have this property because its units are panel-specific: the same λ moves a Gemma-3-27B representation by a much larger ambient distance than it does a Qwen2.5-0.5B one, which is why the legacy protocol must run a per-panel λ-HPO grid and ours does not.

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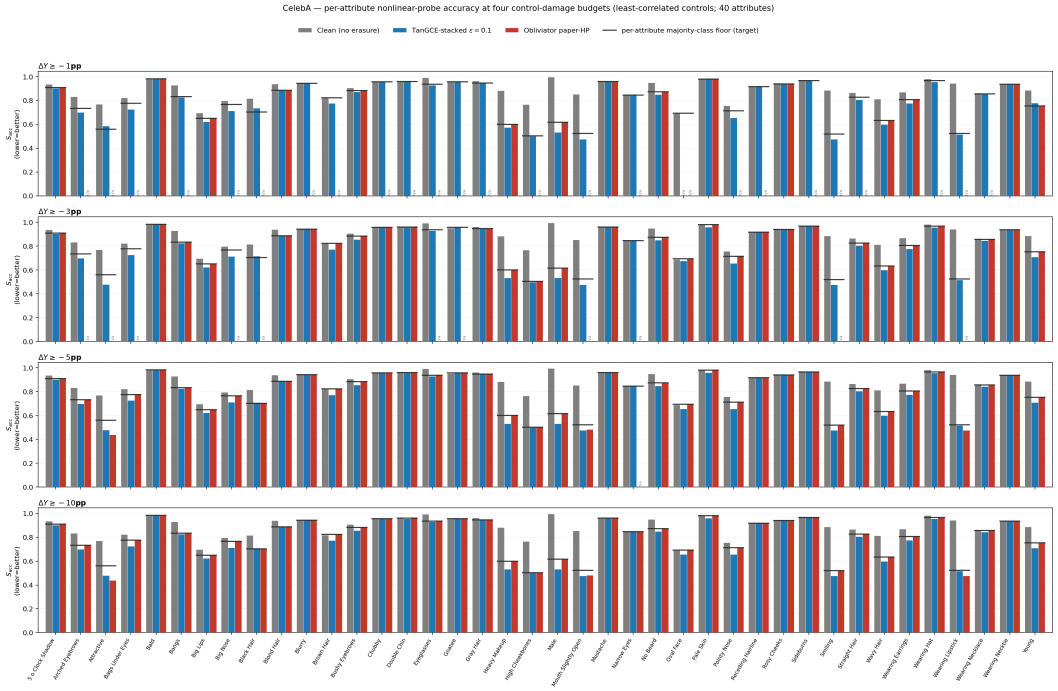


(a) 5 *least*-correlated controls (low entanglement).

(b) 5 *most*-correlated controls (high entanglement).

Figure 5: **CelebA: per-attribute nonlinear-probe accuracy at four control-damage budgets, both control-set regimes.** Same layout as Figures 2–4 but with the 40 CelebA attributes as groups along the x-axis. Each group has three bars — gray = clean, blue = TanGCE-stacked ε=0.1, red = Obliviator paper-HP — and a black horizontal tick at the per-attribute majority-class floor. "n/a" marks attributes where the method had no trajectory step inside the ΔY budget; this is especially common for Obliv in the most-correlated regime, where its first step already exceeds the budget on heavily entangled attributes. TanGCE-stacked stays at or below the per-attribute floor on the great majority of attributes in both regimes.

Table 3: **Phase-1 closed-form ablations on the 39-panel NLP grid (50%-depth layer).** Same fresh nonlinear probe ($h = 128$, 200 SGD steps, patience 3) is trained on each post-erasure representation. *LEACE* is the affine LEACE eraser. *LEACE + CovMatch K=2* additionally projects out the top-2 eigenvectors of $\Delta\Sigma = \Sigma_+ - \Sigma_-$ (rank-3 closed-form erasure total). *TanGCE-stacked* is the proposed full pipeline (Phase 1 + iterative tangent loop, ε=0.1); we report the best $|S-\text{floor}|$ trajectory step subject to $\Delta Y \geq -5$pp. NL = nonlinear probe accuracy on test (lower is better; floor is the per-panel majority class). ΔY in pp on the control task. Bold marks the row's NL minimum.

Model	Concept	Floor (NL*)	LEACE NL	LEACE ΔYpp	LEACE + CovMatch K=2 NL	LEACE + CovMatch K=2 ΔYpp	TanGCE-stacked ε=0.1 NL	TanGCE-stacked ε=0.1 ΔYpp
Qwen2.5-0.5B	Sycophancy	0.550	**0.550**	+0.0	**0.550**	-2.1	**0.550**	-0.1
	Gender	0.539	0.709	-0.1	0.684	-0.4	**0.549**	-4.1
	Safety	0.510	0.752	+0.0	0.529	+0.0	**0.510**	-0.1
Qwen2.5-1.5B	Sycophancy	0.550	**0.550**	-0.1	**0.550**	-1.5	**0.550**	+1.2
	Gender	0.539	0.822	-0.3	0.788	-1.7	**0.592**	-3.9
	Safety	0.510	0.759	+2.9	**0.510**	+2.9	**0.510**	+0.0
Qwen2.5-3B	Sycophancy	0.675	**0.675**	+0.1	**0.675**	-2.7	**0.675**	-0.7
	Gender	0.539	0.857	-1.3	0.783	-0.1	**0.548**	-4.1
	Safety	0.510	0.718	+0.5	0.520	+0.5	**0.510**	+0.0
Gemma-2-2B	Sycophancy	0.543	0.643	+0.0	**0.543**	-0.4	**0.543**	-0.2
	Gender	0.539	0.722	+0.3	0.681	-0.7	**0.539**	-4.9
	Safety	0.510	0.715	+0.7	0.654	+0.7	**0.510**	+0.0
Gemma-2-9B	Sycophancy	0.569	0.751	-0.1	**0.569**	-0.7	0.569	+0.0
	Gender	0.539	0.638	+0.2	0.562	-0.4	**0.539**	-3.5
	Safety	0.510	0.623	+0.0	**0.510**	+0.0	**0.510**	+0.0
Gemma-2-27B	Sycophancy	0.553	0.695	-0.2	**0.553**	-0.9	**0.553**	-0.3
	Gender	0.539	0.663	+0.0	0.605	-2.5	**0.539**	-0.5
	Safety	0.510	0.736	+0.7	0.643	+1.4	**0.510**	+0.0
Llama-3.2-1B	Sycophancy	0.551	0.849	-0.1	**0.551**	-0.2	**0.551**	+0.0
	Gender	0.539	0.855	-0.4	0.738	+0.8	**0.536**	-3.9
	Safety	0.510	0.817	+0.3	0.778	+0.3	**0.498**	+0.0
Llama-3.2-3B	Sycophancy	0.551	0.896	-0.0	**0.551**	+0.0	**0.551**	+0.0
	Gender	0.539	0.889	+0.2	0.794	-0.4	**0.535**	-2.8
	Safety	0.510	0.837	+0.1	0.821	+0.2	**0.526**	+0.0
Mistral-7B	Sycophancy	0.551	0.920	+0.0	**0.551**	-0.1	**0.551**	+0.2
	Gender	0.539	0.851	-0.4	0.815	-0.3	**0.606**	-4.8
	Safety	0.510	0.825	+0.0	0.811	+0.0	**0.532**	+0.2
Gemma-3-1B	Sycophancy	0.551	0.727	-0.1	**0.551**	-2.2	**0.551**	-0.0
	Gender	0.539	0.708	+0.0	0.613	-0.8	**0.539**	+2.0
	Safety	0.510	0.726	+0.0	0.723	-1.0	**0.510**	+1.0
Gemma-3-4B	Sycophancy	0.551	**0.551**	-0.0	**0.551**	-0.9	**0.551**	-0.4
	Gender	0.539	0.704	-0.5	0.580	-2.7	**0.539**	+4.9
	Safety	0.510	0.735	+1.0	0.609	+1.0	**0.510**	-0.2
Gemma-3-12B	Sycophancy	0.556	0.613	+0.0	**0.556**	-0.5	**0.556**	-0.5
	Gender	0.539	**0.539**	+3.5	0.606	-8.7	0.539	+3.1
	Safety	0.510	0.589	+3.8	**0.510**	+3.8	**0.510**	+0.0
Gemma-3-27B	Sycophancy	0.526	0.586	-0.2	0.566	-0.1	**0.526**	+0.0
	Gender	0.539	0.700	+1.7	0.678	-2.6	**0.539**	-4.8
	Safety	0.510	0.686	+0.9	**0.510**	+0.7	**0.510**	+0.0
Mean sycophancy		0.560	0.693	-0.0	0.563	-0.9	**0.560**	-0.1
Mean gender		0.539	0.743	+0.2	0.687	-1.6	**0.549**	-2.1
Mean safety		0.510	0.732	+0.8	0.625	+0.8	**0.512**	+0.1

Table 4: CelebA all-40-attribute summary (CLIP ViT-B/32). The upper block reports methods evaluated under the validation-HPO protocol from the language-model experiments: λ^* is selected per method per concept from the grid $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation NL probe accuracy. The lower block reports trajectory-based methods (TanGCE+LEACE preprocessing and Obliviator) for which we instead pick the trajectory step with lowest probe accuracy subject to mean $\Delta Y \geq -5$pp on the 5 control attributes. NL Leak: mean nonlinear test accuracy across 40 concepts (lower = better). Mean ΔCtrl: mean control accuracy change averaged over 40 concepts. Worst ΔCtrl: worst single control drop averaged over 40 concepts (only available for λ-HPO methods that report per-control values).

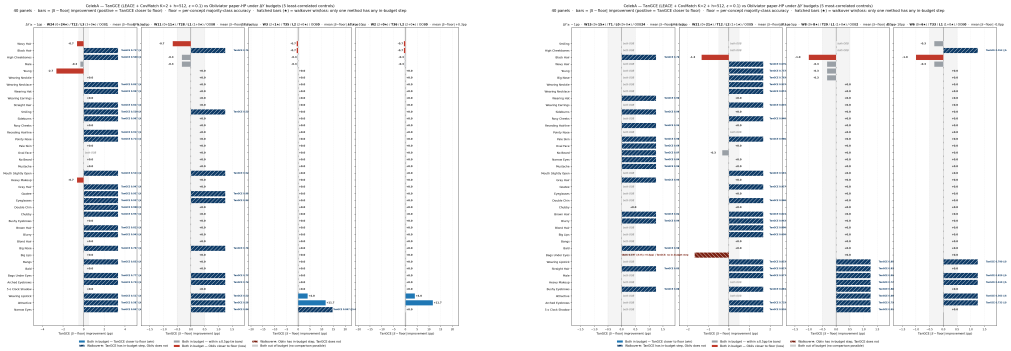
Method	NL Leak↓	Mean ΔCtrl	Worst ΔCtrl	Med. λ^*
λ-HPO selection ($\lambda^* \in \{0.5, 1.0, 2.0, 4.0, 8.0\}$, lowest validation NL):				
Random Global	0.883	-0.4%	-1.6%	2.0
Random Per-Samp.	0.882	-0.3%	$\mathbf{-1.2\%}$	0.5
Linear Proj.	0.869	-0.8%	-2.7%	4.0
INLP	0.859	-3.5%	-6.9%	1.0
LEACE	0.839	$\mathbf{-0.2\%}$	$\mathbf{-1.2\%}$	2.0
AmbGCE	0.875	-0.6%	-2.8%	0.5
TanGCE (canonical, $\alpha = \frac{1}{2}$)	0.707	-0.7%	-2.1%	8.0
Trajectory-based selection (best step at $\Delta Y \geq -5$pp):				
Obliviator	0.799	-2.1%	—	—
TanGCE+LEACE+CovMatch K=2 (ours)	**0.777**	$\mathbf{-1.1\%}$	—	—

Table 5: CLIP ViT-B/32 concept erasure on CelebA (validation HPO for λ). Each method's λ^* is selected from $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation nonlinear probe accuracy; the same probe (MLP hidden 256, 1500 steps, AdamW lr=10^{-3}) is used for both HPO and evaluation. L/NL Leak: retrained linear / nonlinear probe accuracy on test (lower = better). ΔCtrl: mean control-task accuracy change across 5 least-correlated attributes (closer to zero = more surgical). Representation: CLIP ViT-B/32 pooled embedding (768 dimensions).

		Male			Smiling			Blond_Hair		
Method	λ^*	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl
Random Global	0.5	.997	.990	-0.3	.903	.903	-0.3	.937	.923	-0.3
Random Per-Samp.	0.5	.993	.990	-0.1	.867	.887	-1.3	.933	.920	$+0.1$
Linear Proj.	1.0	.987	.987	-0.7	.903	.913	$\mathbf{+0.1}$	.923	.940	-1.4
INLP	1.0	.587	.987	$\mathbf{-0.2}$	.580	.740	$+0.6$	.713	.917	$\mathbf{+0.7}$
LEACE	0.5	.997	.997	-0.3	.503	.820	$+0.4$	.903	.917	-0.1
AmbGCE	0.5	.997	.997	-0.9	.787	.877	$\mathbf{+0.1}$	.910	.923	-1.1
TanGCE	**8.0**	**.800**	**.847**	-1.1	**.487**	**.443**	-0.1	**.693**	**.720**	-1.6

Table 6: **CelebA visual attributes — surgicality under entangled vs. non-entangled controls.** For each of the 40 binary CelebA attributes, we report nonlinear-probe leakage S_{acc} under two control sets: *5 least-correlated* attributes (low entanglement, easier surgicality) and *5 most-correlated* attributes (high entanglement, harder surgicality). S_{acc} is the lowest probe accuracy reached on a trajectory step satisfying mean $\Delta Y \geq -5$pp on the 5 controls. The **Majority** column reports the per-concept majority-class probe-accuracy floor on the test split (the trivial lower bound any informative probe must beat); a method that drives S_{acc} to this value has fully erased the concept under the control budget. Both methods use `paper-HP` hyperparameters from their respective references; TanGCE adds the Phase-1 LEACE + rank-2 covariance projection plus $h = 512$, $\varepsilon = 0.1$. Lower is better (closer to Majority). Bold marks the lower of the two methods per row.

		5 least-correlated controls		5 most-correlated controls	
Concept	Majority	TanGCE	Obliv	TanGCE	Obliv
5_o_Clock_Shadow	0.910	**0.903**	0.910	0.903	–
Arched_Eyebrows	0.733	**0.700**	0.733	0.707	–
Attractive	0.560	0.480	**0.440**	0.697	–
Bags_Under_Eyes	0.777	**0.727**	0.777	**0.763**	0.777
Bald	0.983	0.983	0.983	0.983	0.983
Bangs	0.833	**0.827**	0.833	0.833	0.833
Big_Lips	0.650	**0.623**	0.650	**0.617**	0.650
Big_Nose	0.767	**0.713**	0.767	**0.680**	0.767
Black_Hair	0.703	0.710	**0.703**	0.713	**0.703**
Blond_Hair	0.887	0.887	0.887	0.887	0.887
Blurry	0.943	0.943	0.943	0.943	0.943
Brown_Hair	0.823	**0.773**	0.823	**0.793**	0.823
Bushy_Eyebrows	0.883	**0.857**	0.883	0.873	–
Chubby	0.957	0.957	0.957	0.957	0.957
Double_Chin	0.960	**0.957**	0.960	**0.950**	0.960
Eyeglasses	0.937	**0.930**	0.937	**0.910**	0.937
Goatee	0.957	0.957	0.957	0.957	0.957
Gray_Hair	0.947	0.947	0.947	0.947	0.947
Heavy_Makeup	0.600	**0.533**	0.600	0.777	–
High_Cheekbones	0.503	**0.500**	0.503	–	–
Male	0.617	**0.533**	0.617	0.953	–
Mouth_Slightly_Open	0.523	**0.477**	0.483	**0.477**	0.523
Mustache	0.960	0.960	0.960	0.960	0.960
Narrow_Eyes	0.847	0.847	–	0.847	0.847
No_Beard	0.873	**0.850**	0.873	**0.870**	0.873
Oval_Face	0.693	**0.657**	0.693	**0.660**	0.693
Pale_Skin	0.980	**0.960**	0.980	**0.973**	0.980
Pointy_Nose	0.713	**0.657**	0.713	**0.657**	0.713
Receding_Hairline	0.917	0.917	0.917	**0.913**	0.917
Rosy_Cheeks	0.940	0.940	0.940	0.940	0.940
Sideburns	0.967	0.967	0.967	0.967	0.967
Smiling	0.520	**0.477**	0.520	–	–
Straight_Hair	0.827	**0.807**	0.827	0.807	–
Wavy_Hair	0.633	**0.600**	0.633	**0.607**	0.633
Wearing_Earrings	0.807	**0.777**	0.807	**0.770**	0.807
Wearing_Hat	0.967	**0.957**	0.967	**0.960**	0.967
Wearing_Lipstick	0.523	0.517	**0.477**	0.853	–
Wearing_Necklace	0.857	**0.843**	0.857	0.857	0.857
Wearing_Necktie	0.937	0.937	0.937	0.937	0.937
Young	0.753	**0.710**	0.753	**0.697**	0.753
Mean	0.804	**0.782**	0.798	**0.831**	0.850



(a) 5 least-correlated controls (low entanglement). (b) 5 most-correlated controls (high entanglement).

Figure 6: **CelebA per-attribute floor-distance win-margin under control-damage budgets.** Solid blue/red bars show $\left(|S^{\text{Obliv}} - S_{\text{floor}}| - |S^{\text{TanGCE}} - S_{\text{floor}}|\right) \times 100$ on panels where *both* methods have at least one trajectory step within the ΔY budget; positive (blue) = TanGCE closer to the per-concept majority-class floor. **Hatched dark-blue bars (⋆)** mark *walkover* panels where TanGCE has at least one in-budget step but Obliv has *none* — its very first step already exceeds the control-damage budget. Hatched dark-red mirrors the (rare) opposite case. Light-grey bars are panels where neither method has an in-budget step. Methods: TanGCE = the proposed pipeline (LEACE + rank-2 $\Delta\Sigma$ + closed-form λ_i at fixed ε=0.1, §5.3.4); Obliv = Obliviator paper-HP. Tie band ± 0.5pp shaded gray. Per-subplot title reports W/T/L counts (in-budget) plus walkover wins⋆.

Table 7: λ sweep comparing AmbGCE and TanGCE at fixed step sizes across three concept domains (Qwen2.5-0.5B, layer 8). NL: retrained nonlinear MLP probe accuracy (hidden 256, 1500 steps, AdamW lr=10^{-3}; same architecture used for HPO and evaluation) on the target concept (lower = stronger erasure). Both methods use the same GCE concept scorer (hidden 256, 1500 steps). AmbGCE becomes less effective at high λ under the current protocol, while TanGCE continues improving. We interpret this as supportive evidence for the local manifold motivation of tangent-constrained updates, not as a complete characterization of the leakage-utility frontier. The expanded HPO grid $\{0.25, \ldots, 32.0\}$ used in the multi-model experiments further tests this range. Bold: best NL per domain.

		NL↓ at $\lambda =$		
Concept	Method	0.5	4.0	8.0
Sycophancy	AmbGCE	.678	.694	.734
	TanGCE	.662	.596	**.588**
Gender	AmbGCE	.666	.700	.702
	TanGCE	.704	.580	**.470**
Safety	AmbGCE	.664	.776	.770
	TanGCE	.748	.680	**.646**

Table 8: Full erasure method comparison at layer 8 across three concept domains (Qwen2.5-0.5B), plus a Majority Class reference floor. Each λ-based method's λ^* is selected from $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation nonlinear probe accuracy; the same probe architecture (MLP hidden 256, 1500 steps, AdamW lr=10^{-3}) is used for both HPO and final evaluation, ensuring no ranking distortion from mismatched capacity. IGBP and linear baselines use their standard procedure. Majority Class is not an erasure method; it reports the label-imbalance floor implied by the test split. L/NL Leak: retrained linear logistic-regression / nonlinear MLP probe accuracy on the target concept (lower = stronger erasure). ΔCtrl: downstream control-task accuracy change relative to clean representations (closer to zero = more surgical). Control tasks: answer matching (sycophancy), profession classification (gender), helpfulness (safety).

	Sycophancy			Gender (Bias in Bios)			Safety		
Method	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl
Random Global	.685	.748	.000	.719	.758	.000	.798	.805	+.003
Random Per-Samp.	.621	.745	.000	.723	.743	+.003	.784	.799	−.010
Linear Proj.	.683	.740	**.000**	.722	.747	+.001	.807	.800	**−.003**
INLP	.683	.734	**.000**	.722	.737	+.001	.807	.795	−.004
LEACE	.681	.681	**.000**	.719	.739	−.002	.801	.785	−.015
IGBP	.683	.748	−.011	.660	.675	−.015	.777	.759	+.062
AmbGCE	.714	.703	**.000**	.663	.701	**.000**	.780	.778	−.040
TanGCE	**.628**	**.602**	−.015	**.603**	**.618**	−.008	**.654**	**.663**	+.039

Table 12: Per-layer breakdown of TanGCE vs. AmbGCE across models and concepts. Each model is evaluated at three layers corresponding to 25%, 50%, and 75% of its total depth. NL: nonlinear probe accuracy (lower = stronger erasure). ΔCtrl: downstream accuracy change (closer to zero = more surgical). λ^* selected from $\{0.25, 0.5, 1.0, 2.0, 4.0, 8.0, 16.0, 32.0\}$.

			Sycophancy		Gender		Safety	
Model	Method	Layer	NL↓	ΔCtrl	NL↓	ΔCtrl	NL↓	ΔCtrl
Qwen2.5-0.5B	AmbGCE	6	0.704	+0.000	0.724	+0.001	0.765	−0.043
	TanGCE	6	**0.592**	+0.000	**0.564**	−0.037	**0.669**	+0.022
	AmbGCE	12	0.781	+0.000	0.715	−0.002	0.806	−0.037
	TanGCE	12	**0.724**	+0.000	**0.538**	−0.060	**0.713**	−0.002
	AmbGCE	18	0.933	+0.000	0.870	−0.008	0.847	−0.042
	TanGCE	18	**0.838**	−0.000	**0.634**	−0.017	**0.753**	+0.002
Qwen2.5-1.5B	AmbGCE	7	0.731	+0.000	0.733	−0.002	0.781	−0.010
	TanGCE	7	**0.672**	+0.000	**0.560**	−0.070	**0.691**	+0.021
	AmbGCE	14	0.798	+0.000	0.858	−0.002	0.826	−0.011
	TanGCE	14	**0.675**	−0.007	**0.577**	−0.053	**0.692**	−0.077
	AmbGCE	21	0.907	+0.000	0.935	−0.001	0.865	−0.047
	TanGCE	21	**0.765**	−0.001	**0.658**	−0.005	**0.769**	−0.073
Qwen2.5-3B	AmbGCE	9	0.808	+0.000	0.855	−0.007	0.787	−0.002
	TanGCE	9	**0.752**	−0.011	**0.550**	−0.018	**0.672**	−0.024
	AmbGCE	18	0.821	+0.000	0.870	−0.001	0.823	−0.017
	TanGCE	18	**0.761**	+0.000	**0.585**	−0.040	**0.698**	−0.056
	AmbGCE	27	0.953	+0.000	0.930	+0.002	0.895	−0.002
	TanGCE	27	**0.906**	+0.000	**0.634**	−0.018	**0.839**	−0.095
Gemma-2-2B	AmbGCE	6	0.628	+0.000	0.750	+0.000	0.718	−0.015
	TanGCE	6	**0.603**	+0.000	**0.545**	−0.010	**0.550**	+0.001
	AmbGCE	13	0.668	+0.000	0.795	+0.003	0.760	−0.010
	TanGCE	13	**0.632**	+0.000	**0.532**	−0.000	**0.572**	−0.017
	AmbGCE	20	0.673	+0.000	0.885	+0.003	0.757	−0.019
	TanGCE	20	**0.641**	+0.000	**0.642**	−0.025	**0.610**	−0.004
Gemma-2-9B	AmbGCE	10	0.686	+0.000	0.813	−0.005	0.801	−0.006
	TanGCE	10	**0.684**	−0.000	**0.535**	−0.034	**0.665**	+0.006
	AmbGCE	21	0.770	+0.000	0.768	+0.001	0.751	−0.024
	TanGCE	21	**0.672**	+0.000	**0.664**	−0.015	**0.664**	+0.000
	AmbGCE	32	0.780	+0.000	0.927	−0.009	0.763	−0.034
	TanGCE	32	**0.696**	+0.000	**0.727**	−0.032	**0.664**	−0.002
Gemma-2-27B	AmbGCE	12	0.812	+0.000	0.890	−0.003	0.809	−0.005
	TanGCE	12	**0.680**	+0.000	**0.655**	−0.057	**0.716**	−0.002
	AmbGCE	23	0.764	+0.000	0.819	−0.003	0.818	−0.098
	TanGCE	23	**0.665**	−0.001	**0.659**	−0.028	**0.685**	−0.011
	AmbGCE	34	0.701	+0.000	0.889	−0.000	0.821	−0.014
	TanGCE	34	**0.670**	+0.000	**0.695**	−0.079	**0.725**	−0.006
Llama-3.2-1B	AmbGCE	4	0.790	+0.000	0.856	−0.001	0.790	−0.028
	TanGCE	4	**0.691**	+0.000	**0.565**	−0.030	**0.612**	+0.008
	AmbGCE	8	0.887	+0.000	0.901	+0.003	0.839	−0.039
	TanGCE	8	**0.704**	+0.000	**0.550**	−0.019	**0.660**	−0.017
	AmbGCE	12	0.815	+0.000	0.935	−0.001	0.853	−0.027
	TanGCE	12	**0.734**	+0.000	**0.755**	−0.012	**0.715**	−0.029
Llama-3.2-3B	AmbGCE	7	0.813	+0.000	0.903	−0.002	0.820	−0.016
	TanGCE	7	**0.728**	−0.001	**0.571**	−0.017	**0.649**	−0.007
	AmbGCE	14	0.909	+0.000	0.929	−0.001	0.860	−0.046
	TanGCE	14	**0.768**	+0.000	**0.610**	−0.010	**0.745**	−0.023
	AmbGCE	21	0.924	+0.000	0.945	−0.005	0.872	−0.027
	TanGCE	21	**0.747**	+0.000	**0.758**	−0.009	**0.733**	−0.030
Mistral-7B	AmbGCE	8	0.820	+0.000	0.895	−0.004	0.852	−0.020
	TanGCE	8	**0.730**	+0.000	**0.599**	−0.012	**0.692**	−0.005
	AmbGCE	16	0.935	+0.000	0.914	−0.003	0.884	−0.032
	TanGCE	16	**0.724**	+0.000	**0.596**	−0.028	**0.730**	−0.038
	AmbGCE	24	0.938	+0.000	0.926	−0.002	0.879	−0.039
	TanGCE	24	**0.721**	−0.001	**0.708**	−0.009	**0.748**	−0.018

Table 13: Supplementary deeper-layer sycophancy comparison at $\lambda = 1.0$ after 7 rounds, across layers 8, 16, and 23. Columns T (tangent-constrained) and A (ambient baseline) compare erasure methods. Utility is answer-matching change relative to clean representations. Leakage and displacement decrease from left to right (lower is better); utility increases from left to right (higher is better). This table is included as supplementary evidence of directional consistency with the layer-8 comparison in Appendix Table 8. Deeper-layer tangent runs continue to reduce leakage more than ambient baselines across this sweep.

Layer	T leak	A leak	T util	A util	T disp	A disp
8	0.063	0.213	-20.7pp	-0.5pp	2.88	0.62
16	0.153	0.400	+0.0pp	+0.0pp	3.79	0.27
23	0.138	0.361	-0.1pp	-0.2pp	44.50	3.11

Table 14: Supplementary deeper-layer safety comparison at $\lambda = 1.0$ after 7 rounds, across layers 8, 16, and 23. Columns T (tangent-constrained) and A (ambient baseline) compare erasure methods. Utility is pairwise helpfulness change. Leakage and displacement decrease from left to right (lower is better); utility increases from left to right (higher is better). This table is included as supplementary evidence of directional consistency with the layer-8 comparison in Appendix Table 8. Deeper-layer tangent runs continue to reduce leakage more than ambient baselines across this sweep.

Layer	T leak	A leak	T util	A util	T disp	A disp
8	0.0725	0.2583	-0.052	-0.023	2.70	0.18
16	0.1292	0.3517	-0.092	+0.003	4.37	0.60
23	0.1483	0.3404	-0.062	-0.152	38.32	5.30

Table 15: Local Table-2-style CelebA visual comparison using CLIP ViT-B/32 embeddings. For each target concept we intervene once per method and evaluate (i) concept leakage via retrained linear/nonlinear probes (lower is better) and (ii) mean control-task accuracy change across 3 least-correlated control attributes (closer to zero is better). Targets and controls: Male $\rightarrow$ Blurry,Narrow_Eyes,Straight_Hair; Smiling $\rightarrow$ Straight_Hair,Bushy_Eyebrows,Gray_Hair; Blond_Hair $\rightarrow$ Straight_Hair,Narrow_Eyes,Blurry. Lambdas follow the layer-8 single-model protocol (Appendix Table 8): all methods λ=0.5 except TanGCE λ=8.0.

	Male			Smiling			Blond_Hair		
Method	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl
Random Global	.993	.994	-0.0pp	.891	.911	+0.0pp	.909	.933	-0.1pp
Random Per-Samp.	.993	.994	+0.0pp	.891	.907	+0.2pp	.907	.932	-0.1pp
Linear Proj.	.993	.992	+0.0pp	.900	.906	-0.3pp	.919	.931	-0.2pp
INLP	.994	.993	-0.0pp	.900	.903	+0.3pp	.918	.932	-0.0pp
LEACE	.994	.991	-0.0pp	.894	.894	+0.2pp	.909	.925	-0.1pp
AmbGCE	.990	.995	-0.2pp	.858	.884	-0.4pp	.893	.920	+0.2pp
TanGCE	.887	.953	-0.8pp	.649	.649	-0.6pp	.699	.747	-2.1pp

Table 16: Per-concept nonlinear probe leakage on CelebA (CLIP ViT-B/32), all 40 binary attributes. Each cell is retrained nonlinear probe test accuracy after erasure (lower = stronger erasure). λ^* is selected per method per concept from $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation NL accuracy. Bold marks the lowest leakage per row. Abbrev.: RandG = Random Global, RandS = Random Per-Sample, LinProj = Linear Projection, AmbGCE = ambient gradient-based concept erasure, TanGCE = tangent gradient-based concept erasure.

Concept	RandG	RandS	LinProj	INLP	LEACE	IGBP	AmbGCE	TanGCE
5_o_Clock_Shadow	.92	.92	.91	.92	.87	.92	.91	**.77**
Arched_Eyebrows	.79	.79	.76	.76	.69	.77	.77	**.59**
Attractive	.73	.74	.69	.73	.63	**.53**	.73	.60
Bags_Under_Eyes	.76	.75	.74	.75	.70	.77	.75	**.69**
Bald	1.00	.99	.99	.99	.98	1.00	.99	**.97**
Bangs	.91	.91	.89	.89	.87	.92	.91	**.68**
Big_Lips	.80	.79	.74	.80	**.68**	.82	.81	.73
Big_Nose	.76	.76	.73	.74	.72	.73	.76	**.61**
Black_Hair	.85	.84	.82	.81	.79	.83	.82	**.46**
Blond_Hair	.92	.92	.94	.92	.92	.92	.92	**.72**
Blurry	.98	.98	.98	.96	.97	.98	.97	**.88**
Brown_Hair	.80	.80	.76	.75	.75	.80	.80	**.52**
Bushy_Eyebrows	.87	.88	.86	.85	.76	.87	.86	**.66**
Chubby	.94	.94	.95	.95	.92	.95	.94	**.81**
Double_Chin	.94	.94	.94	.93	.92	.94	.95	**.85**
Eyeglasses	.99	1.00	.96	.93	.96	1.00	.96	**.90**
Goatee	.96	.97	.97	.93	.96	.96	.95	**.85**
Gray_Hair	.97	.97	.97	.97	.97	.97	.98	**.89**
Heavy_Makeup	.89	.89	.89	.86	.85	.88	.88	**.70**
High_Cheekbones	.79	.79	.77	.75	.70	.57	.76	**.50**
Male	.99	.99	.99	.99	1.00	.98	1.00	**.85**
Mouth_Slightly_Open	.89	.89	.89	.71	.72	.66	.80	**.43**
Mustache	.94	.95	.95	.94	.92	.95	.93	**.77**
Narrow_Eyes	.88	.88	.80	.87	.80	.88	.88	**.78**
No_Beard	.94	.95	.94	.93	.92	.94	.95	**.77**
Oval_Face	.70	.71	.69	.68	**.67**	.69	.67	.72
Pale_Skin	.93	.93	.94	.95	.93	.92	.93	**.83**
Pointy_Nose	.75	.72	.71	.69	**.63**	.72	.75	.64
Receding_Hairline	.94	.94	.96	.96	.94	.94	.95	**.69**
Rosy_Cheeks	.93	.93	.93	.94	.91	.94	.93	**.79**
Sideburns	.95	.95	.94	.94	.95	.94	.94	**.82**
Smiling	.90	.89	.91	.74	.82	.59	.88	**.44**
Straight_Hair	.78	.78	.79	.77	.78	.78	.78	**.49**
Wavy_Hair	.78	.76	.77	.77	.78	.77	.78	**.52**
Wearing_Earrings	.90	.89	.86	.88	.85	.89	.88	**.61**
Wearing_Hat	1.00	1.00	.97	.97	.97	1.00	.97	**.94**
Wearing_Lipstick	.90	.90	.90	.89	.85	.88	.92	**.75**
Wearing_Necklace	.84	.85	.80	.80	.77	.85	.83	**.68**
Wearing_Necktie	.93	.93	.92	.92	.91	.92	.93	**.78**
Young	.86	.86	.85	.84	.83	.86	.87	**.60**
Mean	.88	.88	.87	.86	.84	.86	.87	**.71**

Table 17: Per-concept mean control accuracy change ΔCtrl (%) on CelebA, all 40 binary attributes. For each concept, ΔCtrl is averaged over the 5 least-correlated control attributes (closer to zero = more surgical). λ^* selection matches Table 16.

Concept	RandG	RandS	LinProj	INLP	LEACE	IGBP	AmbGCE	TanGCE
5_o_Clock_Shadow	+0.0	+0.1	-0.7	+0.1	+0.0	+0.1	-2.3	-0.2
Arched_Eyebrows	-1.3	-2.3	-1.2	-0.8	-1.1	-2.5	+0.6	-0.6
Attractive	-0.3	-1.0	-1.7	-0.8	-0.3	-0.9	+0.0	-0.8
Bags_Under_Eyes	-3.7	-1.7	-0.3	-7.7	-0.3	-0.3	-0.3	-0.1
Bald	+0.1	+0.0	-0.1	+0.1	-0.1	+1.1	+2.3	+0.2
Bangs	-3.7	-1.2	-0.5	-7.9	-0.1	-0.7	-0.3	-0.8
Big_Lips	+0.1	-0.0	-0.3	+0.0	-0.1	-0.5	+0.1	-0.1
Big_Nose	-0.1	+0.3	-3.5	-9.7	+0.0	-0.3	-1.8	+0.1
Black_Hair	-0.1	-0.0	+0.1	+0.2	+0.3	-0.3	-0.3	-0.7
Blond_Hair	-0.3	+0.1	-1.4	+0.7	-0.1	-0.4	-1.1	-1.6
Blurry	-1.3	+0.0	-0.1	-5.5	+0.1	-0.7	+0.7	-0.1
Brown_Hair	+0.1	-0.9	-0.3	-6.0	-0.7	+0.4	+0.1	-0.2
Bushy_Eyebrows	+0.0	+0.1	-0.1	-1.5	+0.2	-0.9	-0.3	-0.5
Chubby	-0.1	+0.3	-0.2	-1.4	-0.2	-0.2	+0.4	-2.0
Double_Chin	+0.1	+0.1	-0.4	-12.7	+0.3	+0.7	-0.7	-2.5
Eyeglasses	-0.1	+0.1	-0.0	-0.1	-0.2	-0.4	-0.2	+0.2
Goatee	+0.1	-0.1	+0.0	+0.2	+0.1	+0.5	-0.3	-0.4
Gray_Hair	-0.2	-0.3	-2.1	+0.4	-0.1	-1.1	-1.5	-1.5
Heavy_Makeup	-0.3	-0.1	-1.1	-1.7	-1.3	-1.3	-1.1	-0.9
High_Cheekbones	+0.1	+0.2	+0.1	-0.3	-0.1	-0.4	-0.0	-0.7
Male	-0.3	-0.1	-0.7	-0.2	-0.3	-2.2	-0.9	-1.1
Mouth_Slightly_Open	-0.3	-0.6	+0.1	+0.4	+0.2	+0.3	+0.1	-0.2
Mustache	+0.0	-0.1	+0.1	-10.3	-0.5	+1.0	+0.5	-1.5
Narrow_Eyes	-0.1	-0.2	-0.3	+0.2	-0.1	+0.3	+0.0	+0.0
No_Beard	-1.2	-0.2	-4.0	-6.5	-1.6	-0.1	-0.7	-0.5
Oval_Face	-2.0	+0.1	-0.1	-14.4	-0.6	-0.3	+0.2	-0.3
Pale_Skin	-1.2	-0.1	-0.1	-6.8	-0.9	-0.1	+0.7	-1.0
Pointy_Nose	-0.1	-0.2	+0.3	-1.1	+0.2	-0.7	-0.1	-0.1
Receding_Hairline	-0.1	-0.1	-0.1	-5.9	+0.0	-0.3	+0.1	-1.1
Rosy_Cheeks	-0.5	+0.1	-0.6	-0.1	-0.2	-0.1	-0.1	-1.3
Sideburns	-0.4	+0.0	-0.1	+0.1	+0.1	-0.6	-2.5	-0.1
Smiling	-0.3	-1.3	+0.1	+0.6	+0.4	+0.1	+0.1	-0.1
Straight_Hair	-0.1	-0.2	-0.1	-0.9	-0.1	-0.8	+0.5	-0.7
Wavy_Hair	+0.1	-0.7	-0.9	+0.2	+0.1	-0.4	-0.1	-0.7
Wearing_Earrings	+0.8	-0.3	-1.6	-0.3	-0.1	-1.3	-0.5	-0.3
Wearing_Hat	-0.1	+0.0	-0.1	-0.6	+0.0	+1.0	+0.1	-0.2
Wearing_Lipstick	+0.2	-0.1	-6.1	-21.5	-1.9	-1.2	-1.4	-0.9
Wearing_Necklace	+0.1	-1.7	-1.7	-9.3	+0.3	-1.8	+0.2	-1.2
Wearing_Necktie	+0.0	+0.1	+0.0	+0.2	-0.2	-0.7	-13.4	-1.3
Young	-0.0	-0.7	-1.7	-10.7	+0.1	-0.9	-2.2	-0.8
Mean	-0.4	-0.3	-0.8	-3.5	-0.2	-0.4	-0.6	-0.7

Table 18: Geometry evidence map organizing the appendix diagnostics into four research blocks. Each block addresses a distinct aspect of manifold structure: subspace sufficiency examines whether task-relevant computation concentrates in a low-dimensional linear subspace, perturbation sensitivity tests whether on-manifold directions matter more than orthogonal directions, representation connectivity evaluates whether the manifold remains connected across domains, and multi-layer accumulation tracks error compounding through the network. These diagnostics support the mechanism discussion and motivate the need for manifold-aware erasure methods but do not by themselves establish causality.

Experiment Block	Research Question	Primary Evidence
Subspace sufficiency	Subspace concentration	Top-1 agreement under intrinsic projection
Perturbation sensitivity	High-variance vs. orthogonal directions	KL and top-1 under on/off-subspace noise
Representation connectivity	Shared vs. domain manifolds	Cross-domain interpolation smoothness
Multi-layer accumulation	Error compounding across the network	Simultaneous intrinsic projection across layers